SIGNAL HARBOR · DEVELOPMENT CYCLE

# Monthly performance against benchmark for the family council

One new project run end to end in Signal Harbor: from Mei Chen's own words to a live pipeline, an approved dashboard and delivery, with every screen captured as it happened.

| 21 | about 45 | 4 | 0.011 |
|---|---|---|---|
| steps | minutes, first words to delivery live | people | model cost (USD) |

Evergreen Office is a synthetic family office. All data, documents and people are synthetic. Testing mode (low-cost models) on a private test environment, 11 Oct 2026.

# The project in one paragraph

Every month the family council wants one page showing how the liquid portfolio EG-LIQUID did against the family benchmark EG-OFFICE-BLEND: portfolio return, benchmark return and the difference in points, ready by the 5th business day so the council can decide whether to rebalance. Today Mei Chen builds it by hand in Excel over two days. In this walkthrough Mei described the need to North. Daniel Foo signed the intent and Mei handed it to Leo Park. Span the architect drafted the design pack and Anita Rao approved it. Span the builder ran it on a copy and checked it against September's issued report (4.17%, 2.71%, 1.46 points): every line matched. A corrected pack version 2 was approved and rerun, and the pipeline went live. Lumen proposed a starter dashboard; Daniel approved it and it was set up for Consume and a monthly council email. Two things did not work: Mei's own tile (the difference) could not be added, and the approved dashboard does not appear in Consume. The fifteen findings at the end say what to fix.

# The people

| Person | Role in this project | Touch points |
|---|---|---|
| Mei Chen | Business owner: describes the need and the sources, provides samples and the benchmark report, shapes the dashboard, sets up delivery | 9 |
| Daniel Foo | Sponsor: signs the intent, approves the dashboard and each delivery channel | 4 |
| Anita Rao | Head of data: approves the design pack (and the plan drawn from it) | 3 |
| Leo Park | Data engineer: confirms systems, answers the few questions, marks the samples, builds and takes the pipeline live | 11 |

# The sources

| File | What it holds | How it arrives |
|---|---|---|
| administrator_positions_2026-09-30.xlsx | Harbourline Fund Services month-end positions and market values, 30 Sep 2026 (29 positions; EG-LIQUID 41,200,000) | Upload, monthly, around the 2nd business day |
| index_levels_2026-08_09.csv | Global equity index and global bond index daily levels, 3 Aug to 30 Sep 2026 (equity 1027.17 to 1074.56; bonds 1031.86 to 1041.22) | Upload, monthly |
| northgate_evergreen_growth_I_2026-09-30_final.pdf | Evergreen Growth I statement, final (amended), net asset value 10,180,000 | Email, uploaded monthly |
| bluewater_evergreen_income_II_2026-09-30_final.pdf | Evergreen Income II statement, final, 6,400,000 | Email, uploaded monthly |
| kestrel_macro_2026-09-30_estimate.pdf | Kestrel Ridge Macro Fund statement, estimate, 7,320,000 | Email, uploaded monthly |
| saltmarsh_fund_II_2026-09-30_final.pdf | Saltmarsh Infrastructure Fund II statement, 9,950,000 (not yet confirmed) | Email, uploaded monthly |
| monthly_performance_report_2026-09.pdf | Last month's report as issued: portfolio 4.17%, Office blend 2.71%, difference 1.46 pts; legs 42 / 28 / 30% | Upload, as the benchmark to check against |

## Timeline of the cycle

**01** Build landing · 7 s
**02** New project · 9 s
**03** Intent with North · 9 min 30 s
**04** Daniel signs the intent · 18 s
**05** Hand over to the data engineer · 21 s
**06** Architecture: what I already know, confirm the systems · 16 s
**07** Architecture: the few questions · 21 s
**08** Architecture: samples (upload the Excel, mark the PDF fields) · 53 s
**09** Architecture: the design pack and the mapping sheet · 31 s
**10** Anita approves the design pack (and the build plan) · 57 s
**11** Pipelines: connect and run on a copy (automatic) · 13 s
**12** Pipelines: benchmark request and variance · 19 s
**13** Pack version 2: Leo corrects the definitions · 28 s
**14** Approve pack version 2; the builder reruns by itself · 59 s
**15** Go live · 18 s
**16** Lumen's starter dashboard · 20 s
**17** Mei adds one specific tile · 4 min 47 s
**18** Daniel approves the dashboard · 45 s
**19** Delivery: Consume and the council email · 2 min 01 s
**20** Project complete · 20 s
**21** Consume and the Map show the result · 39 s

| # | Step | Who acted | Time | Outcome |
|---|---|---|---|---|
| 1 | Build landing | Mei Chen (business owner) | 7 s | Done |
| 2 | New project | Mei Chen (business owner) | 9 s | Done |
| 3 | Intent with North | Mei Chen (business owner), with North (agent) | 9 min 30 s | Done |
| 4 | Daniel signs the intent | Daniel Foo (sponsor) | 18 s | Done |
| 5 | Hand over to the data engineer | Mei Chen, then Leo Park (data engineer) | 21 s | Done |
| 6 | Architecture: what I already know, confirm the systems | Leo Park (data engineer), with Span the architect (agent) | 16 s | Done |
| 7 | Architecture: the few questions | Leo Park (data engineer), with Span the architect (agent) | 21 s | Done, one answer changed outside the screen |
| 8 | Architecture: samples (upload the Excel, mark the PDF fields) | Leo Park (data engineer), with Span the architect (agent) | 53 s | Done |
| 9 | Architecture: the design pack and the mapping sheet | Leo Park (data engineer), with Span the architect (agent) | 31 s | Done, with two open rows |
| 10 | Anita approves the design pack (and the build plan) | Anita Rao (head of data), Leo Park (data engineer) | 57 s | Done, approved twice |
| 11 | Pipelines: connect and run on a copy (automatic) | Leo Park (data engineer), with Span the builder (agent) | 13 s | Done, checks 0 of 1 |
| 12 | Pipelines: benchmark request and variance | Leo Park (data engineer), with Span the builder (agent) | 19 s | Done |
| 13 | Pack version 2: Leo corrects the definitions | Leo Park (data engineer), as Span the architect's reviewer | 28 s | Done, new version |
| 14 | Approve pack version 2; the builder reruns by itself | Anita Rao (head of data), Leo Park (data engineer), with Span the builder (agent) | 59 s | Done |

| | | | | |
|---|---|---|---|---|
| 15 | Go live | Leo Park (data engineer), with Span the builder (agent) | 18 s | Done |
| 16 | Lumen's starter dashboard | Mei Chen (business owner), with Lumen (agent) | 20 s | Done, with wrong tiles |
| 17 | Mei adds one specific tile | Mei Chen (business owner), with Lumen (agent) | 4 min 47 s | Not done: tile not added |
| 18 | Daniel approves the dashboard | Mei Chen, then Daniel Foo (sponsor) | 45 s | Done, with a wrong warning |
| 19 | Delivery: Consume and the council email | Mei Chen (business owner), Daniel Foo (sponsor) | 2 min 01 s | Done, no email sent yet |
| 20 | Project complete | Mei Chen (business owner) | 20 s | Done |
| 21 | Consume and the Map show the result | Daniel Foo (sponsor), Anita Rao (head of data) | 39 s | Not as intended: dashboard missing from Consume |

# 01 · Build landing

**Mei Chen (business owner)**

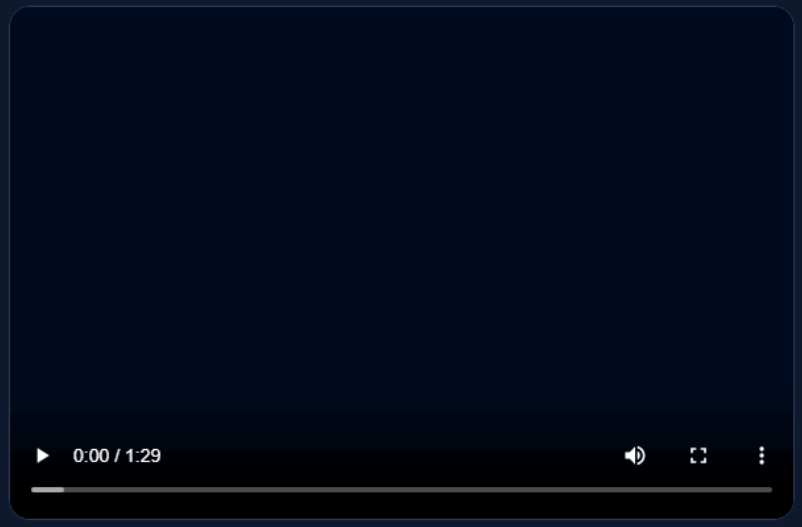
| | |
|---|---|
| **Intended** | Open the Build centre and start the work from the four-step landing page. |
| **What we did** | Signed in as Mei Chen and opened Build. |
| **What happened** | The landing shows the four steps (Intent with North, Architecture with Span, Pipelines with Span, Dashboards and delivery with Lumen), the chapters film and the one project already running (Monthly investment report in USD and SGD, live). |
| **Time taken** | 7 s |
| **If it goes wrong** | Company admin (Grace Ong) checks Mei's access to Build. |

BUILD · SUNDAY 11 OCTOBER

# Build something new, one step at a time

Every new report, check or data product follows the same four steps. You say what you need; agents draft; a person checks and approves each step.

New project




## Chapters

Jump to any part


0:00 **Say what you need**
You talk to North. It writes the brief with you.

0:20 **The data team builds**
Span maps the brief to your data. A data engineer checks it.

0:45 **Check the numbers**
A trial run on a copy, compared with last month's report.

1:05 **Design what people see**
Lumen lays out the views. Someone else approves.

1:20 **Deliver**
Consume, the Morning Note, email or Teams.


STEP 1
### Intent
Say what you need in your own words. North drafts it, the sponsor signs it.
• North

→

STEP 2
### Architecture
Span maps it to the data you have. The data team checks and approves.
• Span

→

STEP 3
### Pipelines
Try it on a copy, check the numbers, then schedule it and go live.
• Span

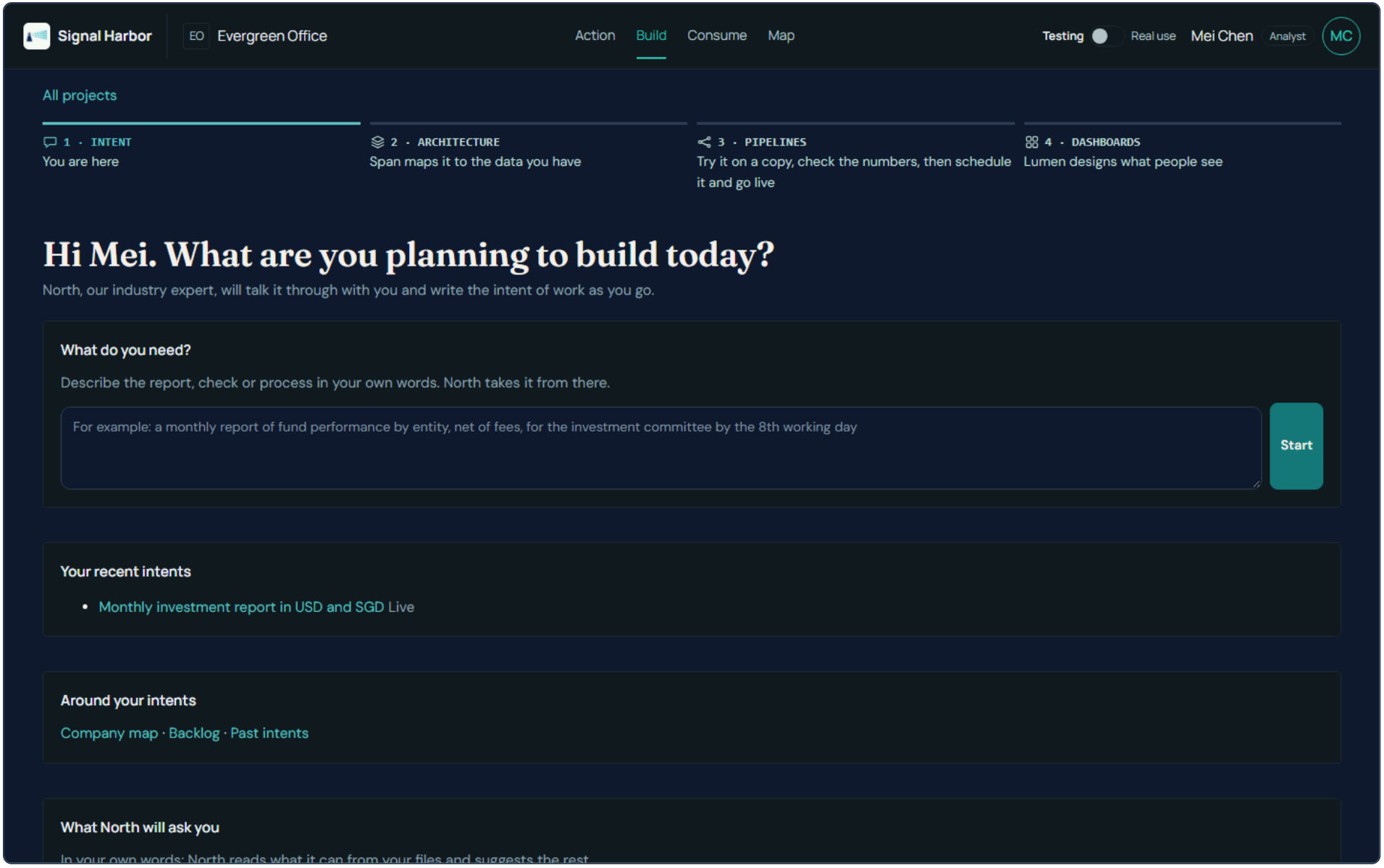
→

STEP 4
### Dashboards and delivery
Lumen designs what people see. Someone else approves, then it is sent.
• Lumen

# 02 · New project

Mei Chen (business owner)

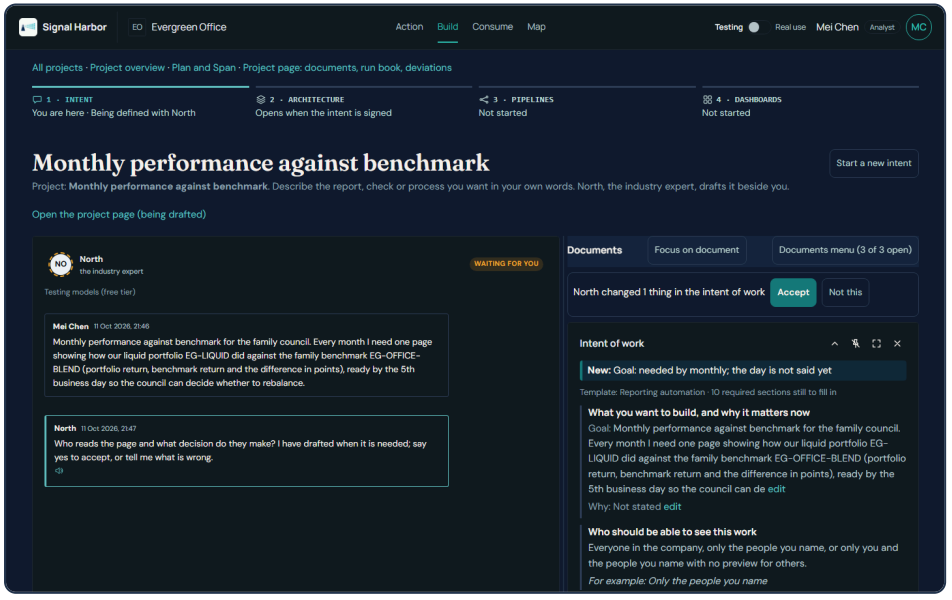
| | |
|---|---|
| Intended | Start a new project from Build. |
| What we did | Clicked New project. |
| What happened | A new project opens on the Intent step with North; Mei is asked to say what she needs in her own words. |
| Time taken | 9 s |
| If it goes wrong | Company admin (Grace Ong) if the button is missing for Mei's role. |

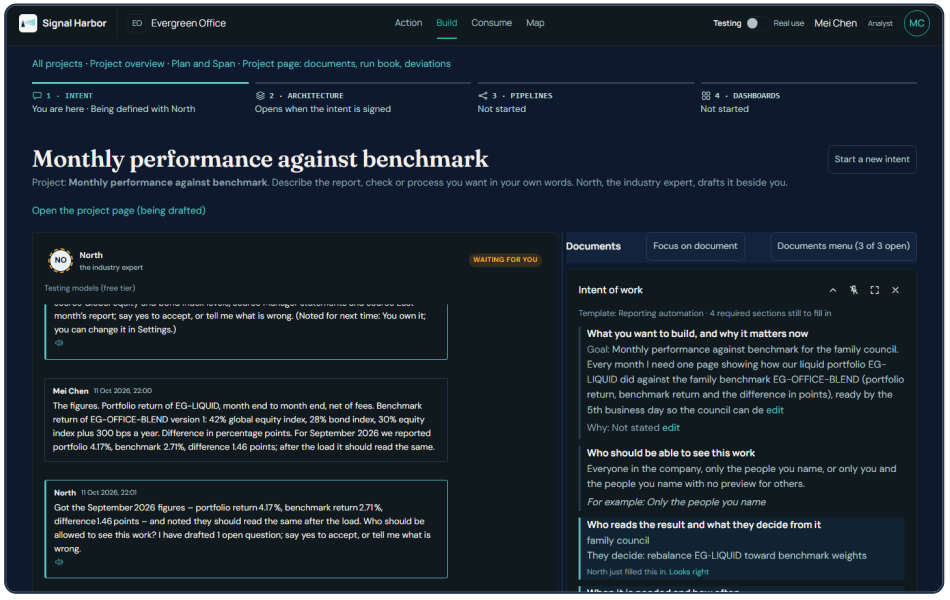

Signal Harbor
EO Evergreen Office
Action Build Consume Map
Testing Real use Mei Chen Analyst MC
All projects
1 · INTENT
You are here
2 · ARCHITECTURE
Span maps it to the data you have
3 · PIPELINES
Try it on a copy, check the numbers, then schedule it and go live
4 · DASHBOARDS
Lumen designs what people see
Hi Mei. What are you planning to build today?
North, our industry expert, will talk it through with you and write the intent of work as you go.
What do you need?
Describe the report, check or process in your own words. North takes it from there.
For example: a monthly report of fund performance by entity, net of fees, for the investment committee by the 8th working day
Start
Your recent intents
Monthly investment report in USD and SGD Live
Around your intents
Company map · Backlog · Past intents
What North will ask you
In your own words: North reads what it can from your files and suggests the rest

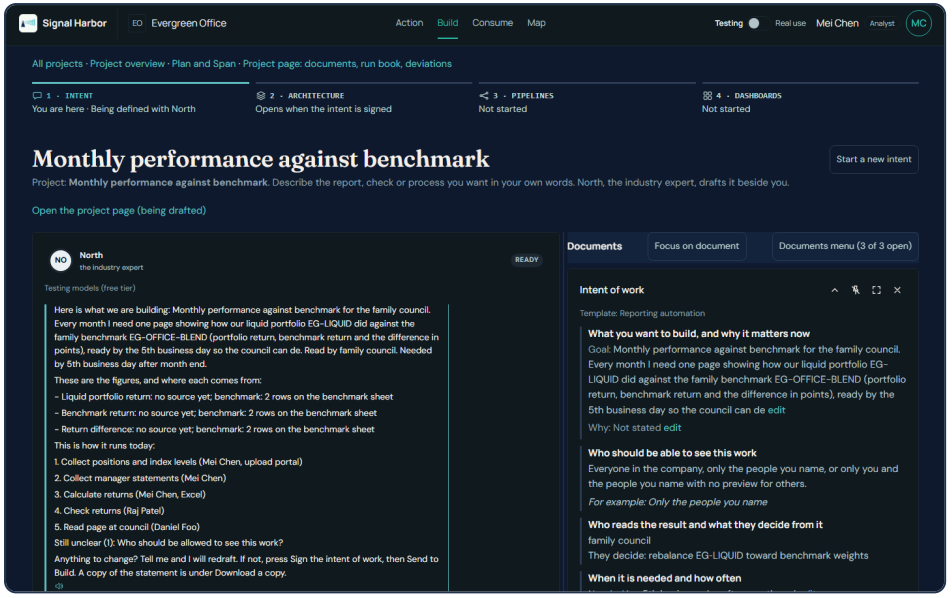
# 03 · Intent with North

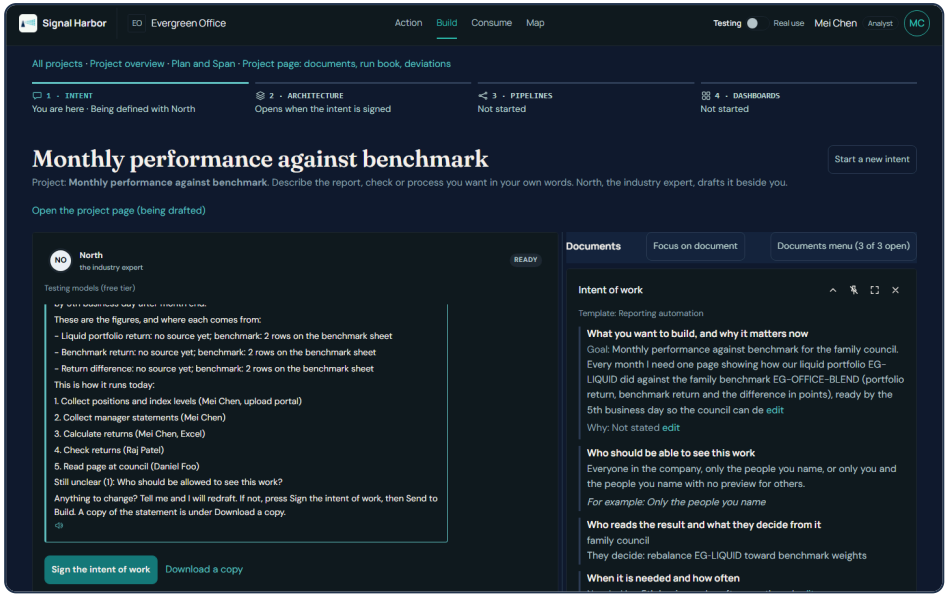
## Mei Chen (business owner), with North (agent)

| | |
|---|---|
| Intended | Mei says in her own words what she needs, which sources feed it and how each arrives; North drafts the intent of work and the process map as they talk. |
| What we did | Ten messages by Mei: the need (EG-LIQUID against EG-OFFICE-BLEND, 5th business day), readers and decision, the four sources with how they arrive (Excel export, index CSV, manager PDFs, last month's report, all by upload), the figures and September's reported values (4.17 / 2.71 / 1.46), today's five steps, access, retention and success; then the result layout, the version 2 open point (Raj Patel) and a request to mark the intent ready. Each North proposal was accepted; the process map was confirmed with That's right. |
| What happened | North asked one question per turn and drafted as it went, e.g. 'Got the September 2026 figures - portfolio return 4.17 %, benchmark return 2.71 %, difference 1.46 points - and noted they should read the same after the load.' It once said 'We still need the version 2 benchmark weights before the pack can be signed'; Mei answered that version 2 is an open point for Raj Patel. After Mei asked, North wrote its closing summary and 'Sign the intent of work' appeared. Gaps: the summary lists each figure as 'no source yet' although Mei named the sources, keeps 'Who should be allowed to see this work?' as unclear although Mei named the four people, and the goal line is cut mid-word ('so the council can de'). |
| Time taken | 9 min 30 s |
| If it goes wrong | Mei corrects any section with North (or the edit link); a wrong figure definition goes to Raj Patel (finance) before signing. |



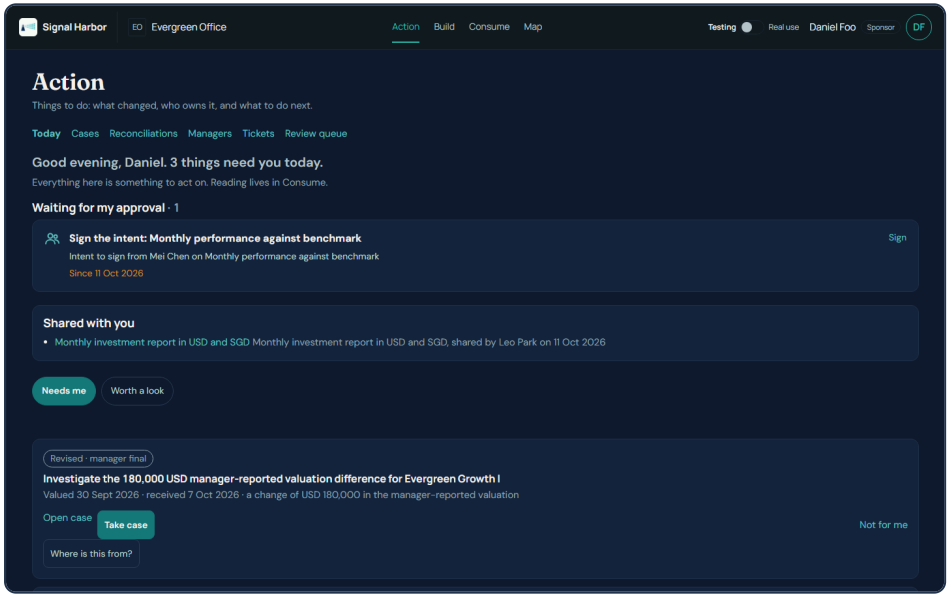

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Testing Real use Mei Chen Analyst MC
All projects · Project overview · Plan and Span · Project page: documents, run book, deviations
1 · INTENT
You are here · Being defined with North
2 · ARCHITECTURE
Opens when the intent is signed
3 · PIPELINES
Not started
4 · DASHBOARDS
Not started
Monthly performance against benchmark
Start a new intent
Project: Monthly performance against benchmark. Describe the report, check or process you want in your own words. North, the industry expert, drafts it beside you.
Open the project page (being drafted)
NO North
the industry expert
READY
Testing models (free tier)
Here is what we are building: Monthly performance against benchmark for the family council. Every month I need one page showing how our liquid portfolio EG-LIQUID did against the family benchmark EG-OFFICE-BLEND (portfolio return, benchmark return and the difference in points), ready by the 5th business day so the council can de. Read by family council. Needed by 5th business day after month end.
These are the figures, and where each comes from:
- Liquid portfolio return: no source yet; benchmark: 2 rows on the benchmark sheet
- Benchmark return: no source yet; benchmark: 2 rows on the benchmark sheet
- Return difference: no source yet; benchmark: 2 rows on the benchmark sheet
This is how it runs today:
1. Collect positions and index levels (Mei Chen, upload portal)
2. Collect manager statements (Mei Chen)
3. Calculate returns (Mei Chen, Excel)
4. Check returns (Raj Patel)
5. Read page at council (Daniel Foo)
Still unclear (1): Who should be allowed to see this work?
Anything to change? Tell me and I will redraft. If not, press Sign the intent of work, then Send to Build. A copy of the statement is under Download a copy.
Documents
Focus on document
Documents menu (3 of 3 open)
Intent of work
Template: Reporting automation
What you want to build, and why it matters now
Goal: Monthly performance against benchmark for the family council. Every month I need one page showing how our liquid portfolio EG-LIQUID did against the family benchmark EG-OFFICE-BLEND (portfolio return, benchmark return and the difference in points), ready by the 5th business day so the council can de edit
Why: Not stated edit
Who should be able to see this work
Everyone in the company, only the people you name, or only you and the people you name with no preview for others.
For example: Only the people you name
Who reads the result and what they decide from it
family council
They decide: rebalance EG-LIQUID toward benchmark weights
When it is needed and how often

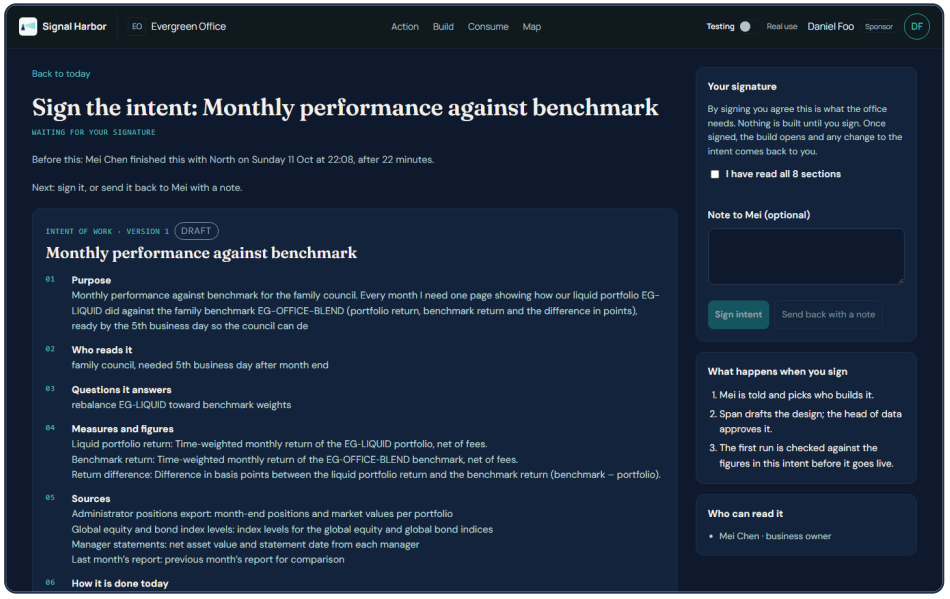

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Testing Real use Mei Chen Analyst MC
All projects · Project overview · Plan and Span · Project page: documents, run book, deviations
1 · INTENT
You are here · Being defined with North
2 · ARCHITECTURE
Opens when the intent is signed
3 · PIPELINES
Not started
4 · DASHBOARDS
Not started
Monthly performance against benchmark
Start a new intent
Project: Monthly performance against benchmark. Describe the report, check or process you want in your own words. North, the industry expert, drafts it beside you.
Open the project page (being drafted)
NO North
the industry expert
READY
Testing models (free tier)
These are the figures, and where each comes from:
- Liquid portfolio return: no source yet; benchmark: 2 rows on the benchmark sheet
- Benchmark return: no source yet; benchmark: 2 rows on the benchmark sheet
- Return difference: no source yet; benchmark: 2 rows on the benchmark sheet
This is how it runs today:
1. Collect positions and index levels (Mei Chen, upload portal)
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3. Calculate returns (Mei Chen, Excel)
4. Check returns (Raj Patel)
5. Read page at council (Daniel Foo)
Still unclear (1): Who should be allowed to see this work?
Anything to change? Tell me and I will redraft. If not, press Sign the intent of work, then Send to Build. A copy of the statement is under Download a copy.
Sign the intent of work
Download a copy
Documents
Focus on document
Documents menu (3 of 3 open)
Intent of work
Template: Reporting automation
What you want to build, and why it matters now
Goal: Monthly performance against benchmark for the family council. Every month I need one page showing how our liquid portfolio EG-LIQUID did against the family benchmark EG-OFFICE-BLEND (portfolio return, benchmark return and the difference in points), ready by the 5th business day so the council can de edit
Why: Not stated edit
Who should be able to see this work
Everyone in the company, only the people you name, or only you and the people you name with no preview for others.
For example: Only the people you name
Who reads the result and what they decide from it
family council
They decide: rebalance EG-LIQUID toward benchmark weights
When it is needed and how often

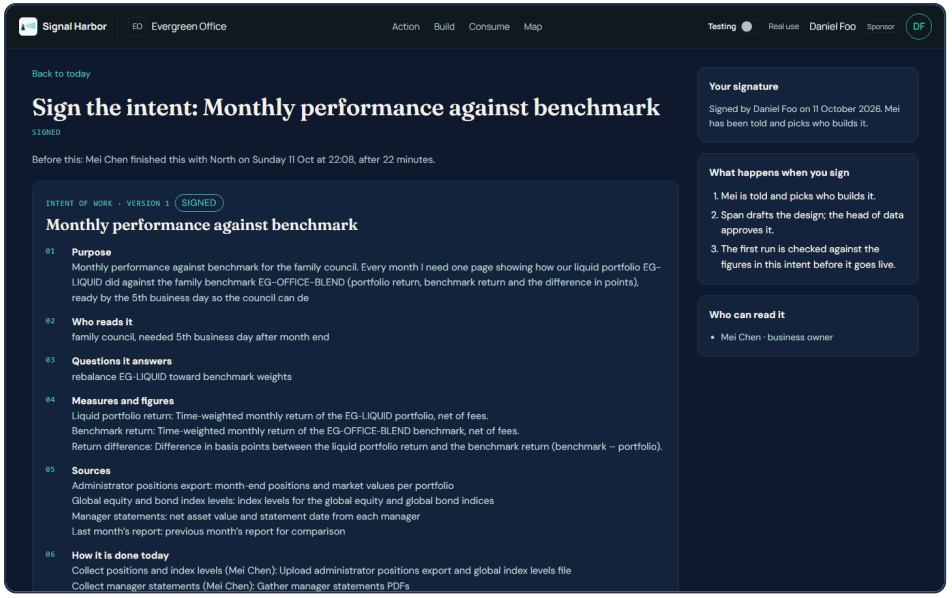
# 04 · Daniel signs the intent

## Daniel Foo (sponsor)

| | |
|---|---|
| Intended | The sponsor reads the intent of work and signs it, so the data team may start. |
| What we did | Daniel opened Action, found the card 'Intent to sign', opened the sign page, ticked that he had read every section and pressed Sign intent. |
| What happened | Action showed the intent to sign. The intent is signed and read-only; Build opens on the project. The signed definition of 'Return difference' reads 'difference in basis points ... (benchmark - portfolio)': North reversed the sign and changed the unit from what Mei said, and the sign page did not flag it. |
| Time taken | 18 s |
| If it goes wrong | Mei Chen reopens the intent with North and fixes the section Daniel queries; Daniel signs again. |



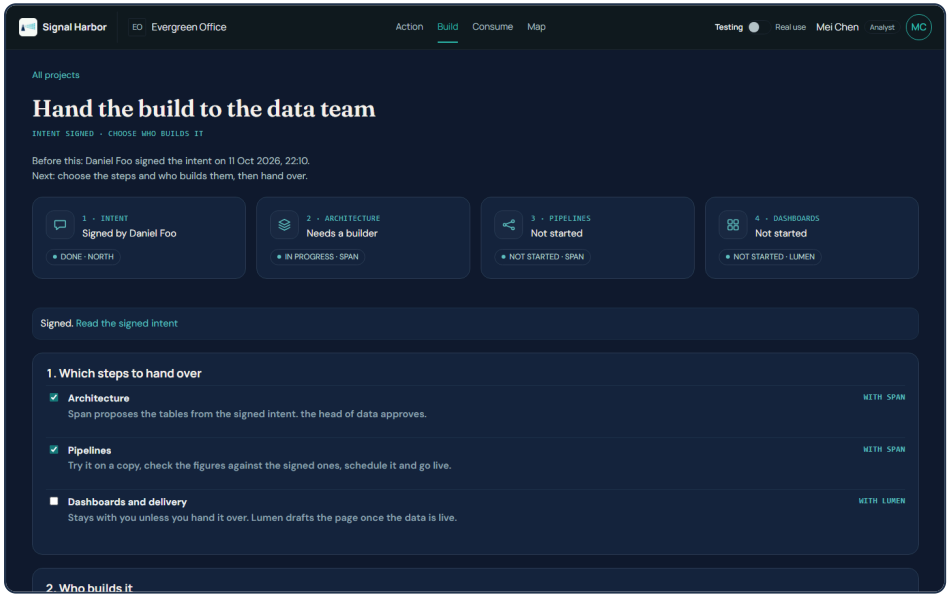



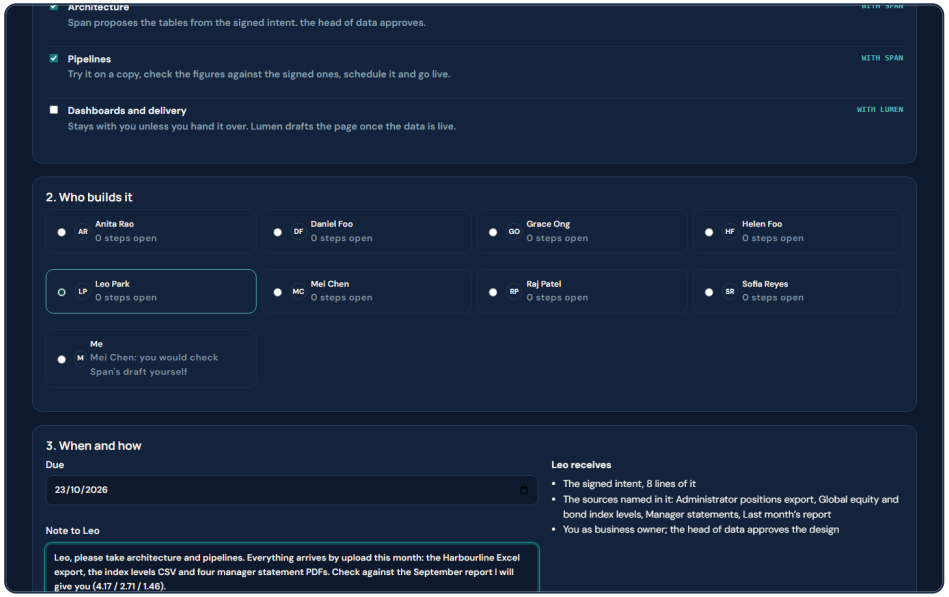

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Testing
Real use
Daniel Foo
Sponsor
DF
Back to today
Sign the intent: Monthly performance against benchmark
SIGNED
Before this: Mei Chen finished this with North on Sunday 11 Oct at 22:08, after 22 minutes.
INTENT OF WORK · VERSION 1 SIGNED
Monthly performance against benchmark
01 Purpose
Monthly performance against benchmark for the family council. Every month I need one page showing how our liquid portfolio EG-LIQUID did against the family benchmark EG-OFFICE-BLEND (portfolio return, benchmark return and the difference in points), ready by the 5th business day so the council can de
02 Who reads it
family council, needed 5th business day after month end
03 Questions it answers
rebalance EG-LIQUID toward benchmark weights
04 Measures and figures
Liquid portfolio return: Time-weighted monthly return of the EG-LIQUID portfolio, net of fees.
Benchmark return: Time-weighted monthly return of the EG-OFFICE-BLEND benchmark, net of fees.
Return difference: Difference in basis points between the liquid portfolio return and the benchmark return (benchmark − portfolio).
05 Sources
Administrator positions export: month-end positions and market values per portfolio
Global equity and bond index levels: index levels for the global equity and global bond indices
Manager statements: net asset value and statement date from each manager
Last month's report: previous month's report for comparison
06 How it is done today
Collect positions and index levels (Mei Chen): Upload administrator positions export and global index levels file
Collect manager statements (Mei Chen): Gather manager statements PDFs
Your signature
Signed by Daniel Foo on 11 October 2026. Mei has been told and picks who builds it.
What happens when you sign
1. Mei is told and picks who builds it.
2. Span drafts the design; the head of data approves it.
3. The first run is checked against the figures in this intent before it goes live.
Who can read it
• Mei Chen · business owner

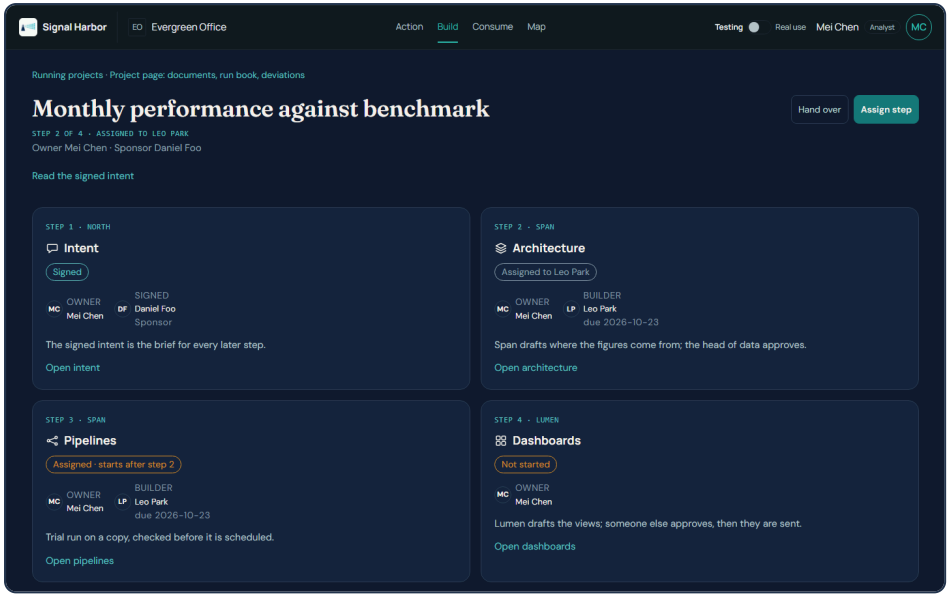
# 05 · Hand over to the data engineer

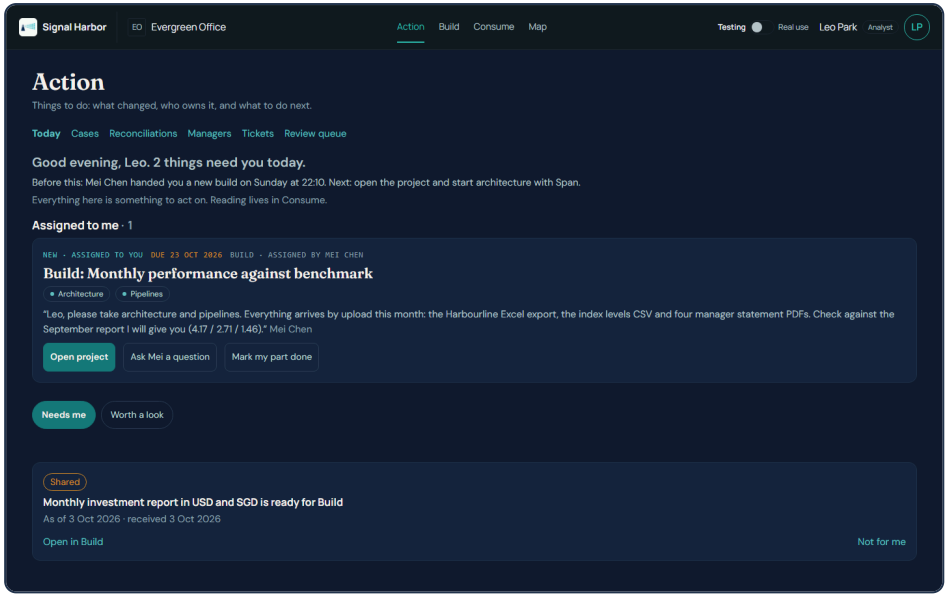
## Mei Chen, then Leo Park (data engineer)

| | |
|---|---|
| Intended | Mei gives Architecture and Pipelines to Leo with a due date and a note; Leo sees the work in his Action list. |
| What we did | Mei picked Leo Park, set due 23 Oct 2026, wrote the note and pressed Hand over. Leo signed in and opened Action. |
| What happened | The project shows Leo as builder; Leo's Action lists the project with its due date. |
| Time taken | 21 s |
| If it goes wrong | Anita Rao (head of data) reassigns the work to another data engineer. |



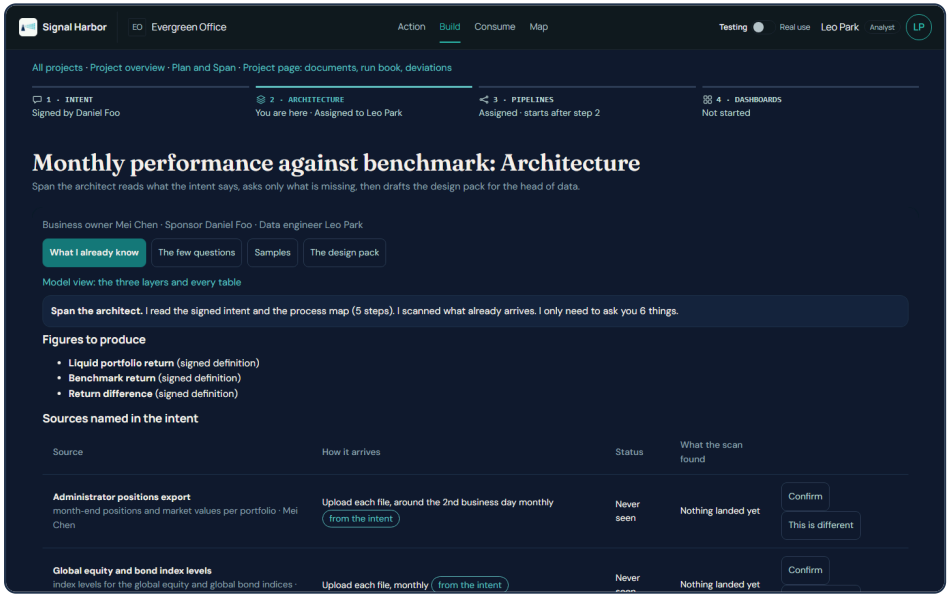

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Testing Real use Mei Chen Analyst MC
Running projects · Project page: documents, run book, deviations
Monthly performance against benchmark
Hand over
Assign step
STEP 2 OF 4 · ASSIGNED TO LEO PARK
Owner Mei Chen · Sponsor Daniel Foo
Read the signed intent
STEP 1 · NORTH
Intent
Signed
MC OWNER Mei Chen
DF SIGNED Daniel Foo Sponsor
The signed intent is the brief for every later step.
Open intent
STEP 2 · SPAN
Architecture
Assigned to Leo Park
MC OWNER Mei Chen
LP BUILDER Leo Park due 2026-10-23
Span drafts where the figures come from; the head of data approves.
Open architecture
STEP 3 · SPAN
Pipelines
Assigned · starts after step 2
MC OWNER Mei Chen
LP BUILDER Leo Park due 2026-10-23
Trial run on a copy, checked before it is scheduled.
Open pipelines
STEP 4 · LUMEN
Dashboards
Not started
MC OWNER Mei Chen
Lumen drafts the views; someone else approves, then they are sent.
Open dashboards

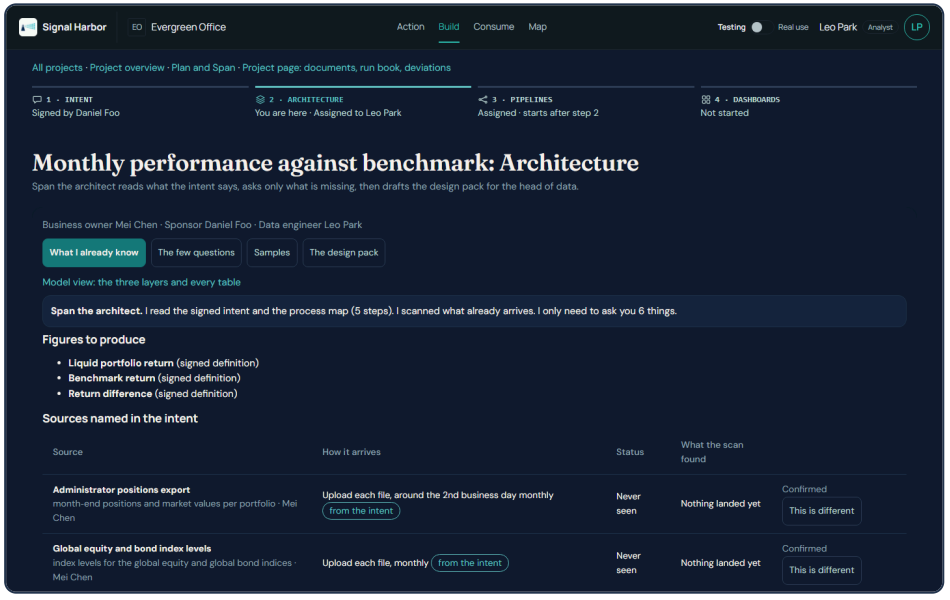

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Testing Real use Leo Park Analyst LP
Action
Things to do: what changed, who owns it, and what to do next.
Today Cases Reconciliations Managers Tickets Review queue
Good evening, Leo. 2 things need you today.
Before this: Mei Chen handed you a new build on Sunday at 22:10. Next: open the project and start architecture with Span.
Everything here is something to act on. Reading lives in Consume.
Assigned to me · 1
NEW · ASSIGNED TO YOU DUE 23 OCT 2026 BUILD · ASSIGNED BY MEI CHEN
Build: Monthly performance against benchmark
Architecture
Pipelines
"Leo, please take architecture and pipelines. Everything arrives by upload this month: the Harbourline Excel export, the index levels CSV and four manager statement PDFs. Check against the September report I will give you (4.17 / 2.71 / 1.46)." Mei Chen
Open project
Ask Mei a question
Mark my part done
Needs me
Worth a look
Shared
Monthly investment report in USD and SGD is ready for Build
As of 3 Oct 2026 · received 3 Oct 2026
Open in Build
Not for me

# 06 · Architecture: what I already know, confirm the systems

Leo Park (data engineer), with Span the architect (agent)

| | |
|---|---|
| Intended | Span reads the signed intent and shows each source as the intent recorded it; Leo confirms rather than re-entering. |
| What we did | Leo opened the project's Architecture step and pressed Confirm on each of the four sources. |
| What happened | Span: 'I read the signed intent and the process map (5 steps). I scanned what already arrives. I only need to ask you 6 things.' The four sources (administrator positions export, index levels, manager statements, last month's report) carry 'from the intent' with how and when they arrive (upload, monthly); the three signed figures are listed. All four confirmed; status 'Never seen', nothing landed yet. Last month's report is listed as a source like the others, not as the benchmark to check against. |
| Time taken | 16 s |
| If it goes wrong | Leo presses 'This is different' and corrects the channel; the intent's system map and the pack change together. Mei answers if the source itself is wrong. |

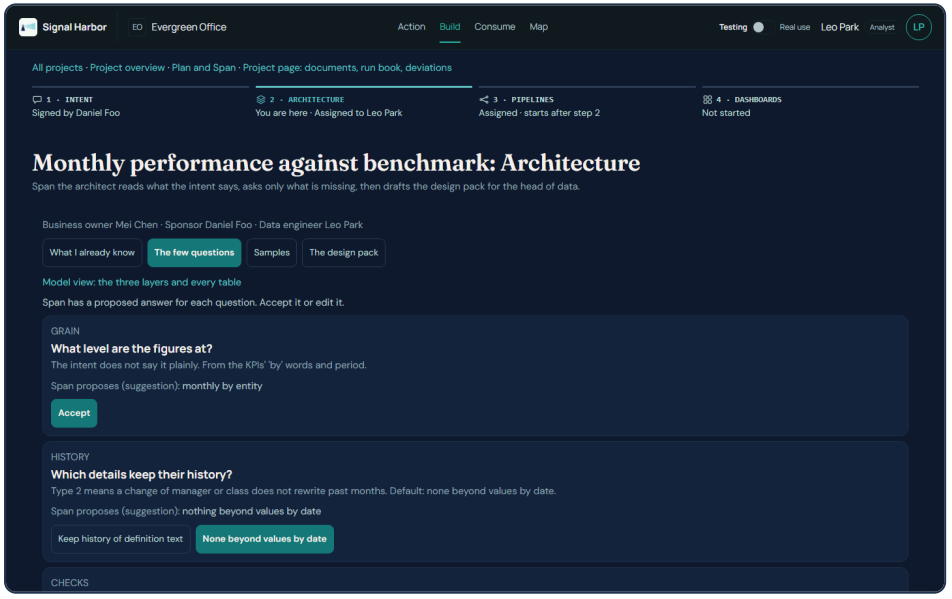


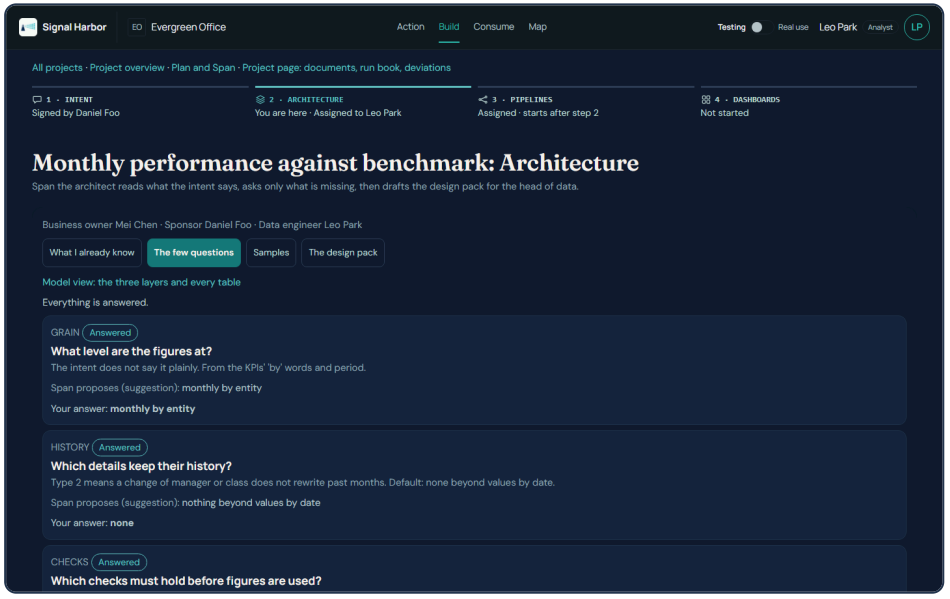



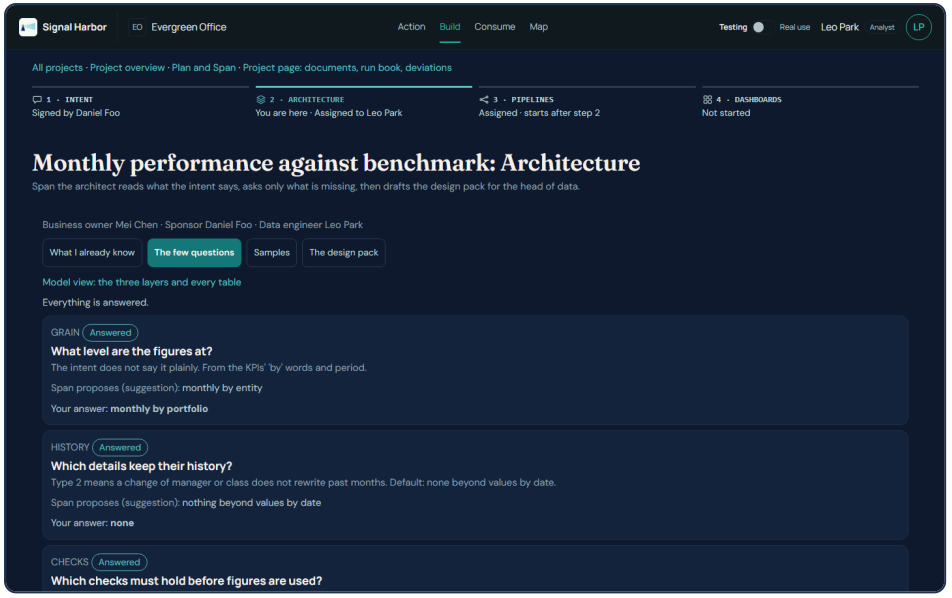
# 07 · Architecture: the few questions

## Leo Park (data engineer), with Span the architect (agent)

| | |
|---|---|
| Intended | Span asks only what the intent and the scan cannot answer, each with a proposed default; Leo accepts or edits. |
| What we did | Span asked four questions (grain, history, checks, freshness). Leo wanted the grain 'monthly by portfolio' instead of Span's 'monthly by entity'; the screen offers only Accept for grain, so he pressed Accept all and continue (history: none beyond values by date; checks: loaded total equals the source footer; freshness: 06:00 monthly, by the 5th business day), then re-answered the grain as 'monthly by portfolio' outside the screen, because the screen has no way to edit it (typing it as plain text was refused: 'That answer does not fit the question.'). |
| What happened | All four answered, grain now by portfolio. Span did not ask again for sources, definitions or access, which the intent already held. The two PDF sources remain open as 'mark the fields' sample questions. |
| Time taken | 21 s |
| If it goes wrong | Leo re-answers a question; Anita Rao (head of data) settles a disputed grain or check at pack approval. |



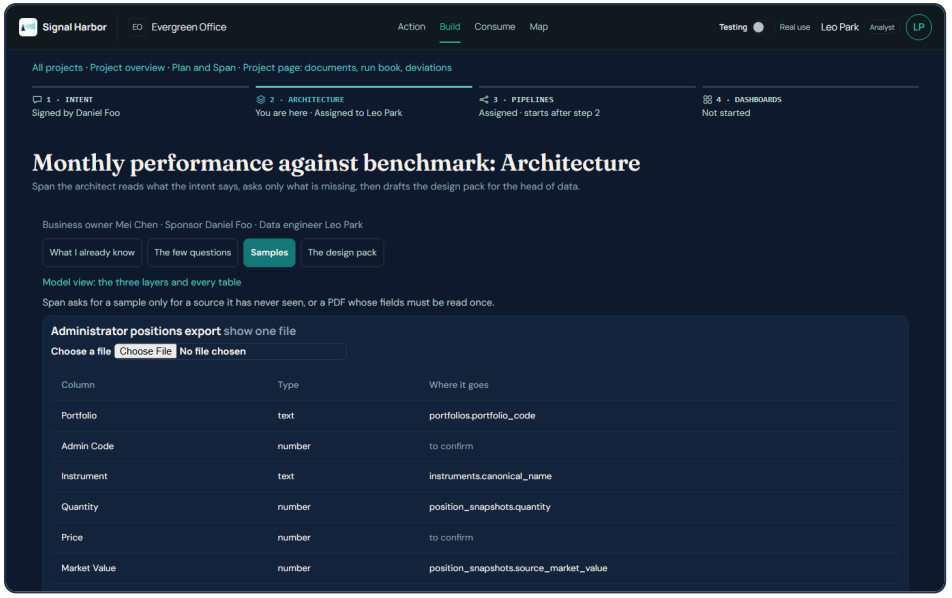




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You are here · Assigned to Leo Park
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Assigned · starts after step 2
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Not started
Monthly performance against benchmark: Architecture
Span the architect reads what the intent says, asks only what is missing, then drafts the design pack for the head of data.
Business owner Mei Chen · Sponsor Daniel Foo · Data engineer Leo Park
What I already know
The few questions
Samples
The design pack
Model view: the three layers and every table
Everything is answered.
GRAIN Answered
What level are the figures at?
The intent does not say it plainly. From the KPIs' 'by' words and period.
Span proposes (suggestion): monthly by entity
Your answer: monthly by portfolio
HISTORY Answered
Which details keep their history?
Type 2 means a change of manager or class does not rewrite past months. Default: none beyond values by date.
Span proposes (suggestion): nothing beyond values by date
Your answer: none
CHECKS Answered
Which checks must hold before figures are used?

# 08 · Architecture: samples (upload the Excel, mark the PDF fields)

Leo Park (data engineer), with Span the architect (agent)

| | |
|---|---|
| Intended | Show Span one file of each never-seen source and mark, once, the fields to read on the PDFs. |
| What we did | Leo chose the files on the Samples tab: the Harbourline positions export (.xlsx) and the index levels (.csv); then the Evergreen Growth I statement (PDF) and marked fund_name, statement_date and net_asset_value on their lines; then last month's report (PDF) and marked portfolio_return, benchmark_return and difference. |
| What happened | Span profiled both tables and proposed where each column goes (e.g. Market Value to position_snapshots.source_market_value, date to market_observations.observed_at); 'to confirm' on Admin Code, Price, Valuation Status, Valuation Date and on both index columns. The statement shows 'Net asset value: 10,180,000' and the marks are listed: 'Span will read these from every statement.' The first PDFs, laid out as two-column tables, could not be marked (each cell became its own line); they were re-issued with one 'label: value' line each. |
| Time taken | 53 s |
| If it goes wrong | Leo re-marks a field; if the statement layout changes, Mei asks the manager for the standard statement; Anita reviews the mapping at approval. |


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Assigned · starts after step 2
4 · DASHBOARDS
Not started
Monthly performance against benchmark: Architecture
Span the architect reads what the intent says, asks only what is missing, then drafts the design pack for the head of data.
Business owner Mei Chen · Sponsor Daniel Foo · Data engineer Leo Park
What I already know
The few questions
Samples
The design pack
Model view: the three layers and every table
Span asks for a sample only for a source it has never seen, or a PDF whose fields must be read once.
Administrator positions export show one file
Choose a file Choose File No file chosen
Column
Type
Where it goes
Portfolio
text
portfolios.portfolio_code
Admin Code
number
to confirm
Instrument
text
instruments.canonical_name
Quantity
number
position_snapshots.quantity
Price
number
to confirm
Market Value
number
position_snapshots.source_market_value


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Assigned · starts after step 2
4 · DASHBOARDS
Not started
Monthly performance against benchmark: Architecture
Span the architect reads what the intent says, asks only what is missing, then drafts the design pack for the head of data.
Business owner Mei Chen · Sponsor Daniel Foo · Data engineer Leo Park
What I already know
The few questions
Samples
The design pack
Model view: the three layers and every table
Span asks for a sample only for a source it has never seen, or a PDF whose fields must be read once.
Administrator positions export show one file
Choose a file Choose File administrator_p...2026-09-30.xlsx
Column
Type
Where it goes
Portfolio
text
portfolios.portfolio_code
Admin Code
number
to confirm
Instrument
text
instruments.canonical_name
Quantity
number
position_snapshots.quantity
Price
number
to confirm
Market Value
number
position_snapshots.source_market_value
Currency
text
position_snapshots.currency
Valuation Status
text
to confirm
Valuation Date
text_date
to confirm
Global equity and bond index levels show one file
Choose a file Choose File index_levels_2026-08_09.csv
Column
Type
Where it goes
date
text_date
market_observations.observed_at
global_equity_index
number
to confirm
global_bond_index
number
to confirm
currency
text
market_observations.base_currency
Manager statements mark the fields to read
Choose a file Choose File No file chosen
Fields marked: fund_name (Fund); statement_date (Statement date); net_asset_value (Net asset value). Span will read these from every statement.
Show the statement lines
Last month's report mark the fields to read
Choose a file Choose File No file chosen
Fields marked: portfolio_return (Portfolio return); benchmark_return (Office blend); difference (bonds). Span will read these from every statement.
Show the statement lines

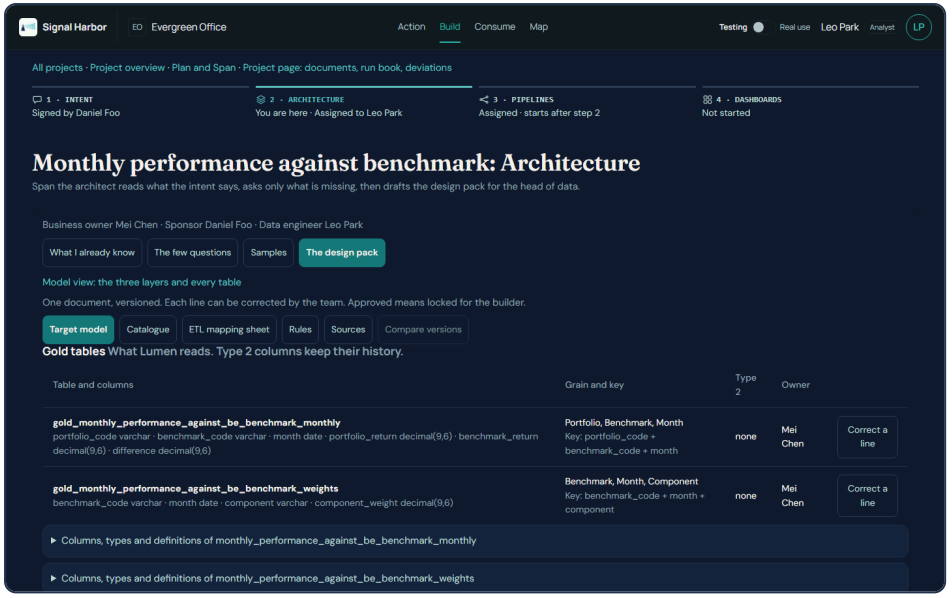

global_bond_index number to confirm
currency text market_observations.base_currency
Manager statements mark the fields to read
Choose a file Choose File northgate_everg..09-30_final.pdf
Fields marked: fund_name (Fund); statement_date (Statement date); net_asset_value (Net asset value). Span will read these from every statement.
Show the statement lines
Name this field net_asset_value
number
5. Statement date: 30 September 2026 Mark here
6. Statement type: Final (amended 7 Oct 2026; replaces the 2 Oct estimate of 10,000,000) Mark here
7. Currency: USD Mark here
8. Net asset value: 10,180,000 Mark here
9. Synthetic document made for the Signal Harbor UC4 walkthrough. Not a real statement. Mark here
Last month's report mark the fields to read
Choose a file Choose File No file chosen
Fields marked: portfolio_return (Portfolio return); benchmark_return (Office blend); difference (bonds). Span will read these from every statement.
Show the statement lines


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Not started
Monthly performance against benchmark: Architecture
Span the architect reads what the intent says, asks only what is missing, then drafts the design pack for the head of data.
Business owner Mei Chen · Sponsor Daniel Foo · Data engineer Leo Park
What I already know
The few questions
Samples
The design pack
Model view: the three layers and every table
Span asks for a sample only for a source it has never seen, or a PDF whose fields must be read once.
Administrator positions export show one file
Choose a file Choose File No file chosen
Column Type Where it goes
Portfolio text portfolios.portfolio_code
Admin Code number to confirm
Instrument text instruments.canonical_name
Quantity number position_snapshots.quantity
Price number to confirm
Market Value number position_snapshots.source_market_value
Currency text position_snapshots.currency
Valuation Status text to confirm
Valuation Date text_date to confirm
Global equity and bond index levels show one file
Choose a file Choose File No file chosen
Column Type Where it goes
date text_date market_observations.observed_at
global_equity_index number to confirm
global_bond_index number to confirm
currency text market_observations.base_currency
Manager statements mark the fields to read
Choose a file Choose File No file chosen
Fields marked: fund_name (Fund); statement_date (Statement date); net_asset_value (Net asset value). Span will read these from every statement.
Show the statement lines
Last month's report mark the fields to read
Choose a file Choose File No file chosen
Fields marked: portfolio_return (Portfolio return); benchmark_return (Office blend); difference (Difference). Span will read these from every statement.
Show the statement lines

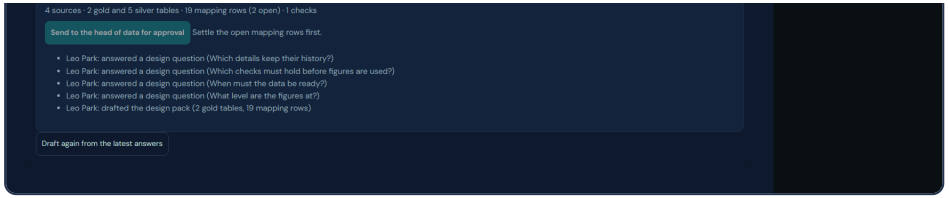
# 09 · Architecture: the design pack and the mapping sheet

Leo Park (data engineer), with Span the architect (agent)

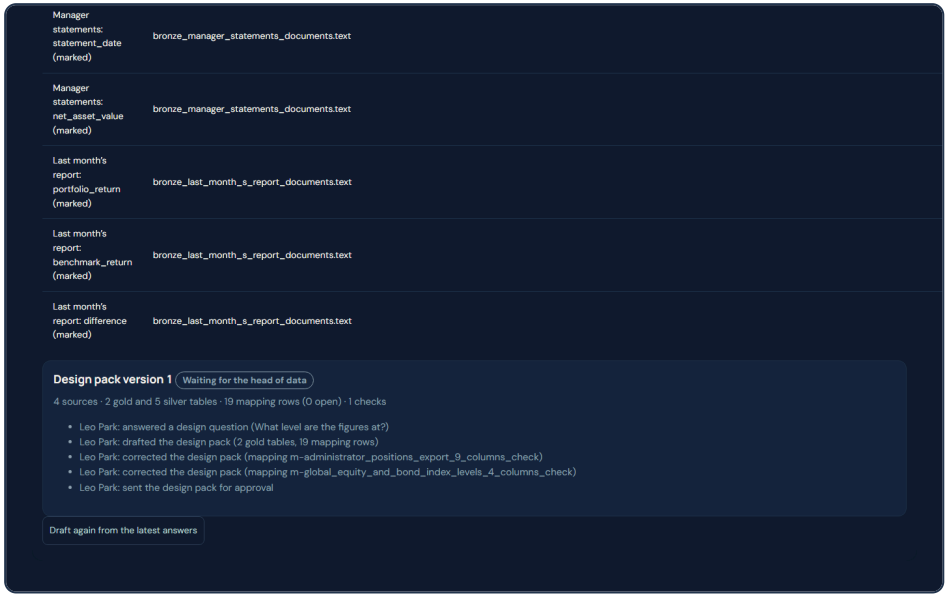
| | |
|---|---|
| Intended | Span drafts one versioned design pack (target model, catalogue, mapping sheet, rules, sources, benchmark check); Leo settles what needs a person and sends it to the head of data. |
| What we did | Leo pressed Draft the design pack, read the Target model, Catalogue, ETL mapping sheet, Rules and Sources tabs, settled the two open mapping rows with a note each, and pressed Send to the head of data for approval. |
| What happened | Pack version 1: 4 sources, 2 gold and 5 silver tables, 19 mapping rows (2 open), 1 check. Gold: returns by portfolio and month (portfolio_return, benchmark_return, difference as DECIMAL(9,6)) and the weight of each benchmark leg; benchmark block EG-LIQUID against EG-OFFICE-BLEND version 1 for 2026-09. The marked PDF fields map to 'Read the value after Net asset value on every statement'. The two open rows were the Excel and CSV samples ('Sample kept; map it before it feeds a table'): the gold returns are computed from the company model (position snapshots, market observations, benchmark components), not from the uploaded files, so Leo settled them in words. Gold table names are cut mid-word ('gold_monthly_performance_against_be_...'). Status: waiting for the head of data. |
| Time taken | 31 s |
| If it goes wrong | Leo corrects a line (a new version); Span redrafts from the latest answers; Anita Rao rejects with a note if the model is wrong. |


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Assigned · starts after step 2
4 · DASHBOARDS
Not started
Monthly performance against benchmark: Architecture
Span the architect reads what the intent says, asks only what is missing, then drafts the design pack for the head of data.
Business owner Mei Chen · Sponsor Daniel Foo · Data engineer Leo Park
What I already know
The few questions
Samples
The design pack
Model view: the three layers and every table
One document, versioned. Each line can be corrected by the team. Approved means locked for the builder.
Target model
Catalogue
ETL mapping sheet
Rules
Sources
Compare versions
Gold tables What Lumen reads. Type 2 columns keep their history.
Table and columns
Grain and key
Type 2
Owner
gold_monthly_performance_against_be_benchmark_monthly
portfolio_code varchar · benchmark_code varchar · month date · portfolio_return decimal(9,6) · benchmark_return decimal(9,6) · difference decimal(9,6)
Portfolio, Benchmark, Month
Key: portfolio_code + benchmark_code + month
none
Mei Chen
Correct a line
gold_monthly_performance_against_be_benchmark_weights
benchmark_code varchar · month date · component varchar · component_weight decimal(9,6)
Benchmark, Month, Component
Key: benchmark_code + month + component
none
Mei Chen
Correct a line
▸ Columns, types and definitions of monthly_performance_against_be_benchmark_monthly
▸ Columns, types and definitions of monthly_performance_against_be_benchmark_weights


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Assigned · starts after step 2
4 · DASHBOARDS
Not started
Monthly performance against benchmark: Architecture
Span the architect reads what the intent says, asks only what is missing, then drafts the design pack for the head of data.
Business owner Mei Chen · Sponsor Daniel Foo · Data engineer Leo Park
What I already know
The few questions
Samples
The design pack
Model view: the three layers and every table
One document, versioned. Each line can be corrected by the team. Approved means locked for the builder.
Target model
Catalogue
ETL mapping sheet
Rules
Sources
Compare versions
2 rows need a person
Source field, then bronze, silver and gold, with the transformation.
Source field
Bronze
Silver
Gold
Transformation or formula
Derived
gold_monthly_performance_against_be_benchmark_monthly.portfolio_code
As it is
Correct a line
Derived
gold_monthly_performance_against_be_benchmark_monthly.benchmark_code
As it is
Correct a line
Derived
gold_monthly_performance_against_be_benchmark_monthly.month
CAST(pr.month AS DATE)
Correct a line
Derived
gold_monthly_performance_against_be_benchmark_monthly.portfolio_return
CAST(pr.portfolio_return AS DECIMAL(9, 6))
Correct a line
Derived
gold_monthly_performance_against_be_benchmark_monthly.benchmark_return
CAST(br.benchmark_return AS DECIMAL(9, 6))
Correct a line
Derived
gold_monthly_performance_against_be_benchmark_monthly.difference
CAST(pr.portfolio_return - br.benchmark_return AS DECIMAL(9, 6))
Correct a line
Derived
gold_monthly_performance_against_be_benchmark_weights.benchmark_code
As it is
Correct a line
Derived
gold_monthly_performance_against_be_benchmark_weights.month
CAST(w.month AS DATE)
Correct a line
Company model:
benchmark_components.market_series_key
gold_monthly_performance_against_be_benchmark_weights.component
c.market_series_key || CASE WHEN COALESCE(c.spread_bps, 0) <> 0 THEN ' plus spread' ELSE '' END
Correct a line
Company model:
benchmark_components.spread_bps
gold_monthly_performance_against_be_benchmark_weights.component
c.market_series_key || CASE WHEN COALESCE(c.spread_bps, 0) <> 0 THEN ' plus spread' ELSE '' END
Correct a line
Company model:
benchmark_components.component_weight
gold_monthly_performance_against_be_benchmark_weights.component_weight
CAST(c.component_weight AS DECIMAL(9, 6))
Correct a line
Administrator positions export: 9 columns
Sample kept; map it before it feeds a table
Map this sample to the model before the builder can use it.
Settle
Global equity and bond index levels: 4 columns
Sample kept; map it before it feeds a table
Map this sample to the model before the builder can use it.
Settle
Manager statements: fund_name (marked)
bronze_manager_statements_documents.text
Read the value after 'Fund' on every statement
Correct a line
Manager statements: statement_date (marked)
bronze_manager_statements_documents.text
Read the value after 'Statement date' on every statement
Correct a line
Manager statements: net_asset_value (marked)
bronze_manager_statements_documents.text
Read the value after 'Net asset value' on every statement
Correct a line
Last month's report: portfolio_return (marked)
bronze_last_month_s_report_documents.text
Read the value after 'Portfolio return' on every statement
Correct a line
Last month's report: benchmark_return (marked)
bronze_last_month_s_report_documents.text
Read the value after 'Office blend' on every statement
Correct a line
Last month's report: difference (marked)
bronze_last_month_s_report_documents.text
Read the value after 'Difference' on every statement
Correct a line
Design pack version 1
Draft


4 sources · 2 gold and 5 silver tables · 19 mapping rows (2 open) · 1 checks
Send to the head of data for approval
Settle the open mapping rows first.
Leo Park: answered a design question (Which details keep their history?)
Leo Park: answered a design question (Which checks must hold before figures are used?)
Leo Park: answered a design question (When must the data be ready?)
Leo Park: answered a design question (What level are the figures at?)
Leo Park: drafted the design pack (2 gold tables, 19 mapping rows)
Draft again from the latest answers

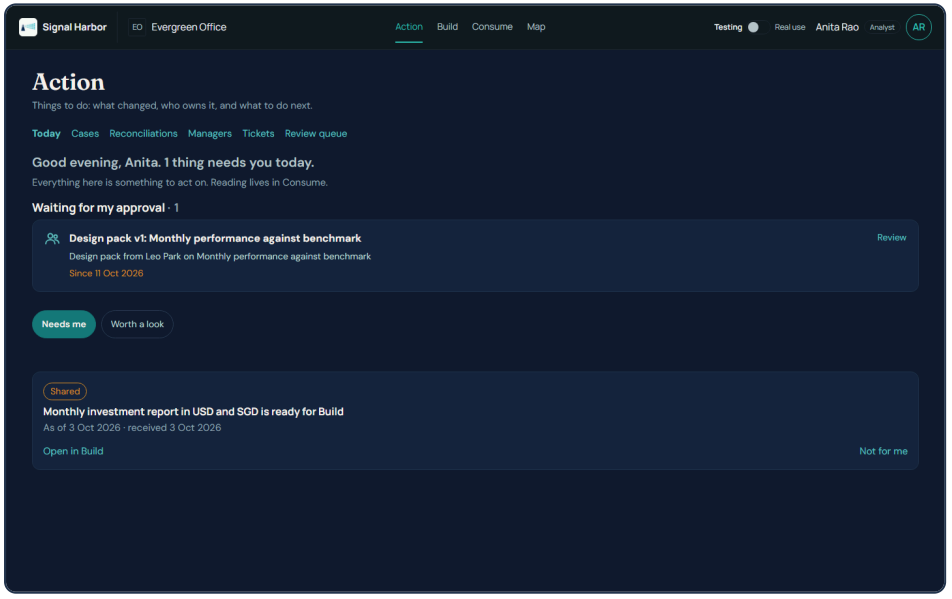

Map this sample to the model before the builder can use it.
Settle
Sample kept; map it before it feeds a table
Map this sample to the model before the builder can use it.
Settle
ger_statements_documents.text
Read the value after 'Fund' on every statement
Correct a line
ger_statements_documents.text
Read the value after 'Statement date' on every statement
Correct a line
ger_statements_documents.text
Read the value after 'Net asset value' on every statement
Correct a line
nonth_s_report_documents.text
Read the value after 'Portfolio return' on every statement
Correct a line
nonth_s_report_documents.text
Read the value after 'Office blend' on every statement
Correct a line
nonth_s_report_documents.text
Read the value after 'Difference' on every statement
Correct a line
ble reads.

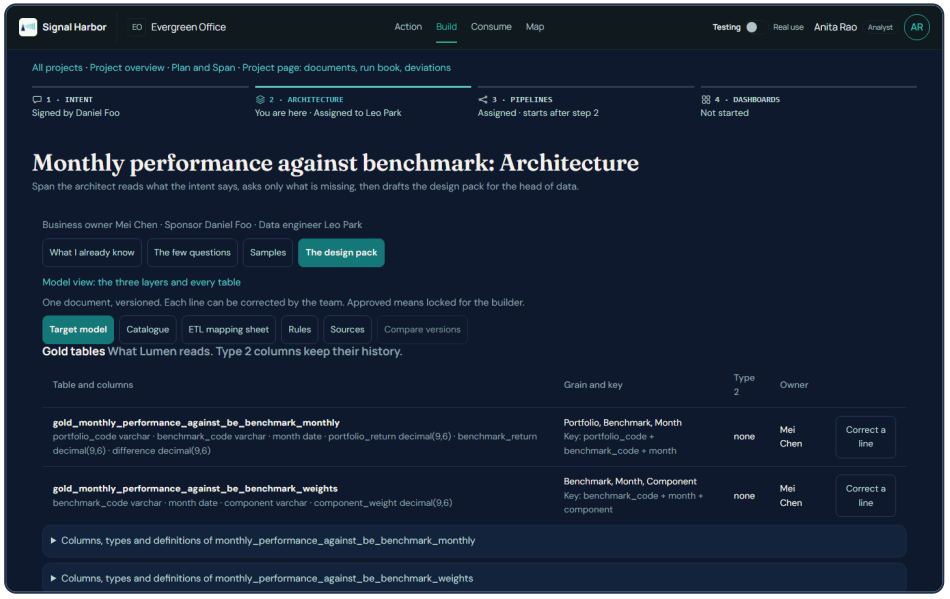

Manager statements: statement_date (marked)
bronze_manager_statements_documents.text
Manager statements: net_asset_value (marked)
bronze_manager_statements_documents.text
Last month's report: portfolio_return (marked)
bronze_last_month_s_report_documents.text
Last month's report: benchmark_return (marked)
bronze_last_month_s_report_documents.text
Last month's report: difference (marked)
bronze_last_month_s_report_documents.text
Design pack version 1
Waiting for the head of data
4 sources · 2 gold and 5 silver tables · 19 mapping rows (0 open) · 1 checks
Leo Park: answered a design question (What level are the figures at?)
Leo Park: drafted the design pack (2 gold tables, 19 mapping rows)
Leo Park: corrected the design pack (mapping m-administrator_positions_export_9_columns_check)
Leo Park: corrected the design pack (mapping m-global_equity_and_bond_index_levels_4_columns_check)
Leo Park: sent the design pack for approval
Draft again from the latest answers

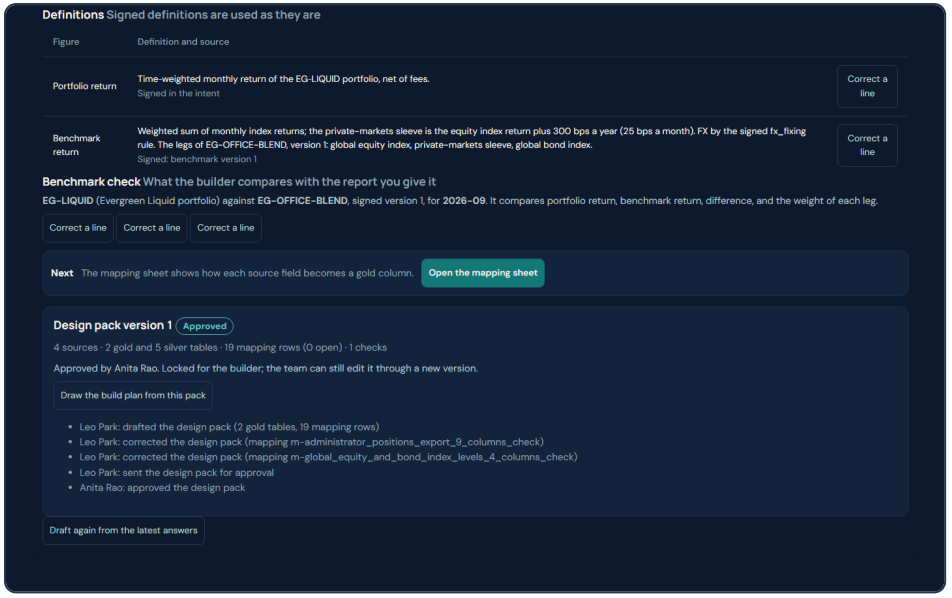
# 10 · Anita approves the design pack (and the build plan)

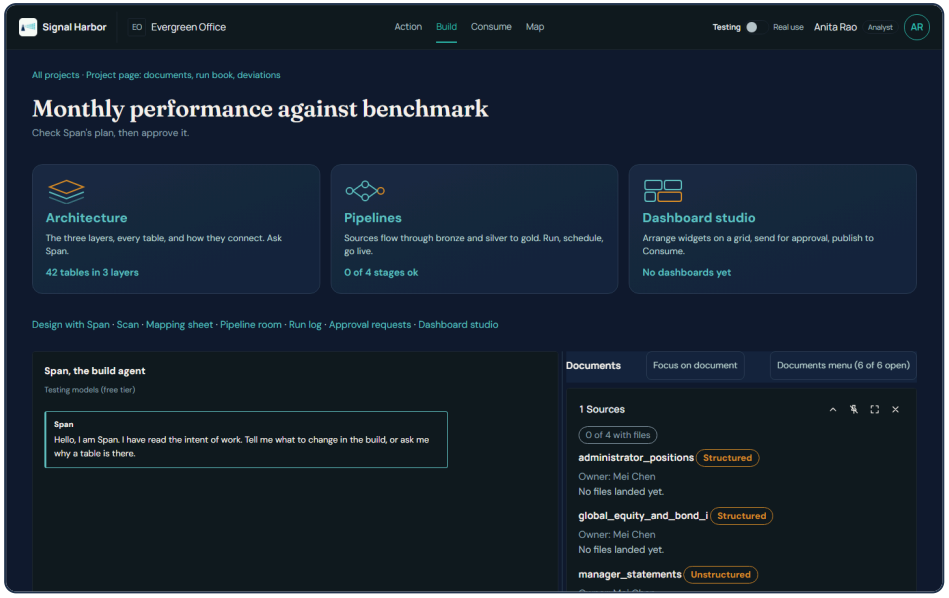
Anita Rao (head of data), Leo Park (data engineer)

| | |
|---|---|
| Intended | The head of data approves the pack; approved means locked for the builder. |
| What we did | Anita opened Action ('Design pack v1: Monthly performance against benchmark', from Leo Park), reviewed the pack and pressed Approve version 1. Leo pressed Draw the build plan from this pack. Anita then approved the plan drawn from it ('the builder may try it on a copy'). |
| What happened | 'Approved by Anita Rao. Locked for the builder; the team can still edit it through a new version.' Plan version 1 drawn from pack version 1 and approved; plan state 'built'. The product asks the head of data to approve twice (the pack, then the plan drawn from it), where the agreed flow has one approval. (The screens for drawing the plan and for approving it repeated earlier captures, so one plan screen is shown.) |
| Time taken | 57 s |
| If it goes wrong | Anita rejects with a note; Leo corrects the pack and resubmits. The maker cannot approve their own version. |



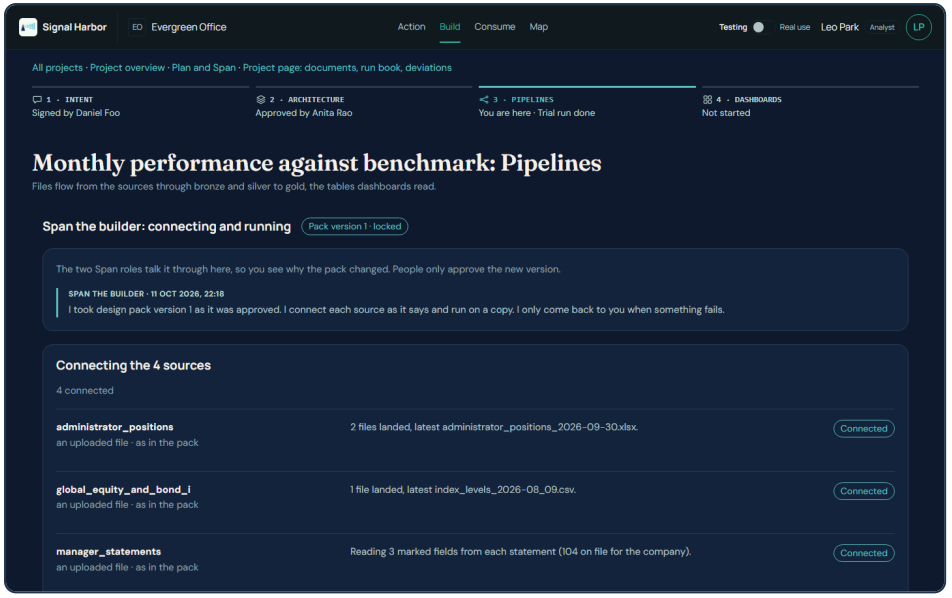

Definitions Signed definitions are used as they are
Figure
Definition and source
Portfolio return
Time-weighted monthly return of the EG-LIQUID portfolio, net of fees.
Signed in the intent
Correct a line
Benchmark return
Weighted sum of monthly index returns; the private-markets sleeve is the equity index return plus 300 bps a year (25 bps a month). FX by the signed fx_fixing rule. The legs of EG-OFFICE-BLEND, version 1: global equity index, private-markets sleeve, global bond index.
Signed: benchmark version 1
Correct a line
Benchmark check What the builder compares with the report you give it
EG-LIQUID (Evergreen Liquid portfolio) against EG-OFFICE-BLEND, signed version 1, for 2026-09. It compares portfolio return, benchmark return, difference, and the weight of each leg.
Correct a line
Correct a line
Correct a line
Next The mapping sheet shows how each source field becomes a gold column.
Open the mapping sheet
Design pack version 1
Approved
4 sources · 2 gold and 5 silver tables · 19 mapping rows (0 open) · 1 checks
Approved by Anita Rao. Locked for the builder; the team can still edit it through a new version.
Draw the build plan from this pack
Leo Park: drafted the design pack (2 gold tables, 19 mapping rows)
Leo Park: corrected the design pack (mapping m-administrator_positions_export_9_columns_check)
Leo Park: corrected the design pack (mapping m-global_equity_and_bond_index_levels_4_columns_check)
Leo Park: sent the design pack for approval
Anita Rao: approved the design pack
Draft again from the latest answers

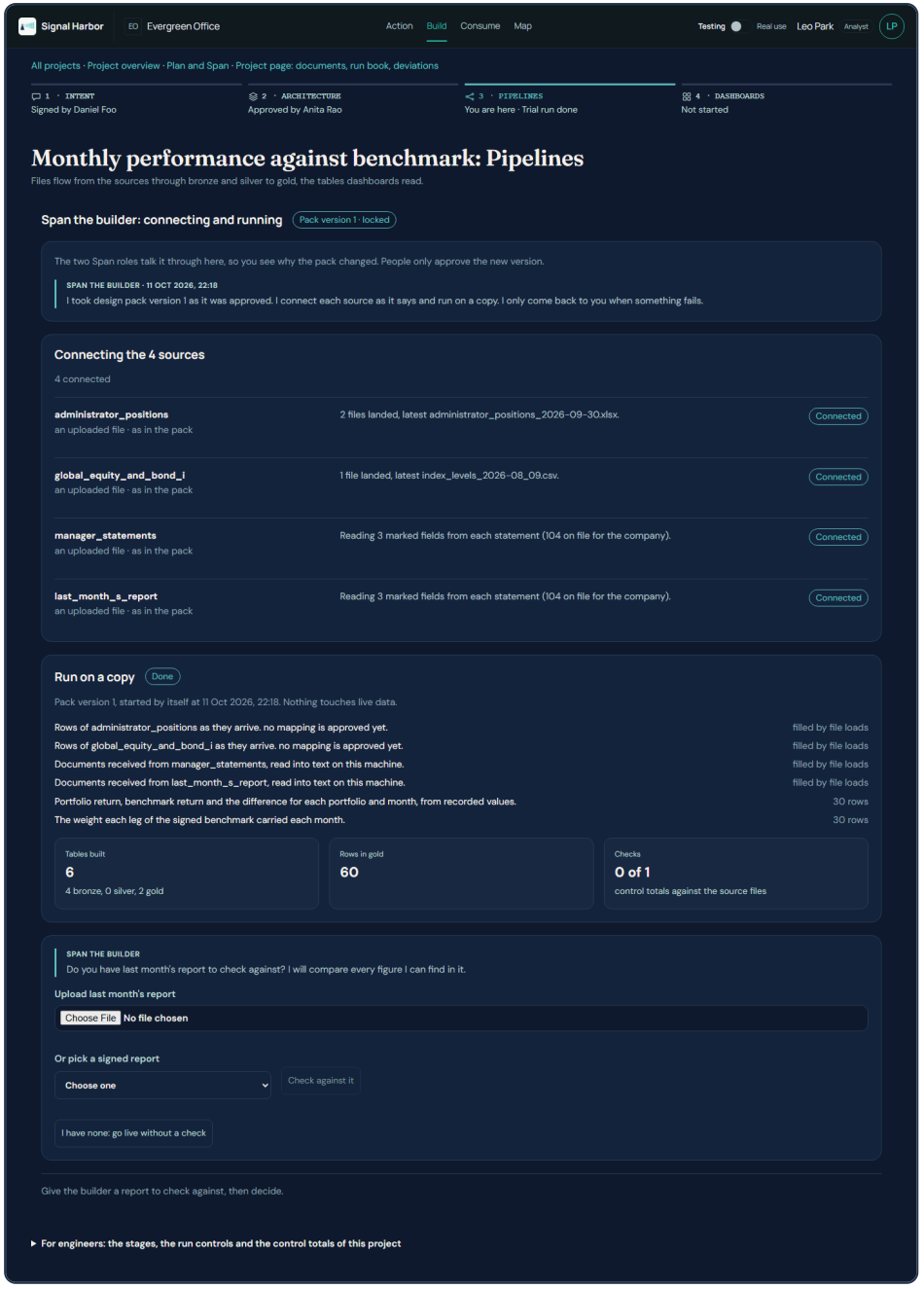

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Action
Build
Consume
Map
Testing
Real use
Anita Rao
Analyst
AR
All projects · Project page: documents, run book, deviations
Monthly performance against benchmark
Check Span's plan, then approve it.
Architecture
The three layers, every table, and how they connect. Ask Span.
42 tables in 3 layers
Pipelines
Sources flow through bronze and silver to gold. Run, schedule, go live.
0 of 4 stages ok
Dashboard studio
Arrange widgets on a grid, send for approval, publish to Consume.
No dashboards yet
Design with Span · Scan · Mapping sheet · Pipeline room · Run log · Approval requests · Dashboard studio
Span, the build agent
Testing models (free tier)
Span
Hello, I am Span. I have read the intent of work. Tell me what to change in the build, or ask me why a table is there.
Documents
Focus on document
Documents menu (6 of 6 open)
1 Sources
0 of 4 with files
administrator_positions
Structured
Owner: Mei Chen
No files landed yet.
global_equity_and_bond_i
Structured
Owner: Mei Chen
No files landed yet.
manager_statements
Unstructured

# 11 · Pipelines: connect and run on a copy (automatic)

Leo Park (data engineer), with Span the builder (agent)

| | |
|---|---|
| Intended | The builder connects each source exactly as the pack says (uploads this month) and runs the pipeline on a copy without waiting for a click. |
| What we did | Leo opened Pipelines. Nothing was clicked: the builder had already started when the plan was approved. |
| What happened | Span the builder: 'I took design pack version 1 as it was approved. I connect each source as it says and run on a copy. I only come back to you when something fails.' 4 of 4 connected (the positions export and index CSV that landed as samples; the marked fields 'from each statement (104 on file for the company)'). Run on a copy: Done, started by itself at 22:18. 6 tables built (4 bronze, 0 silver, 2 gold), 60 gold rows, checks 0 of 1 (control totals against the source files). The bronze tables say 'no mapping is approved yet'. The three other manager statements were not landed: Pipelines has no upload for this month's files, the Architecture sample takes one PDF per source, and the company intake page takes only Excel or CSV. |
| Time taken | 13 s |
| If it goes wrong | The builder comes back to Leo when a connection fails; Mei provides a missing file; Anita decides if a source must change (a new pack version). |

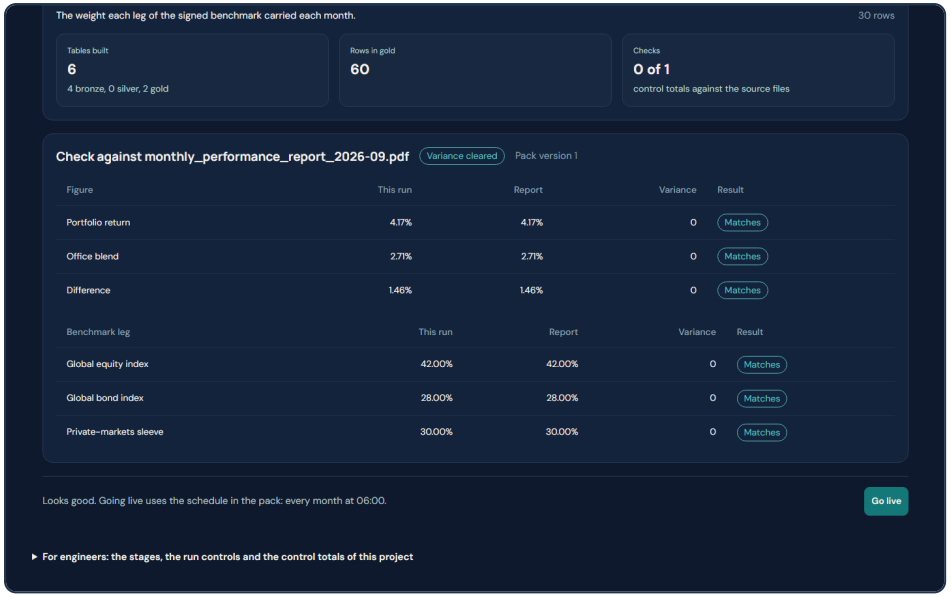

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Action Build Consume Map
Testing Real use Leo Park Analyst LP
All projects · Project overview · Plan and Span · Project page: documents, run book, deviations
1 · INTENT
Signed by Daniel Foo
2 · ARCHITECTURE
Approved by Anita Rao
3 · PIPELINES
You are here · Trial run done
4 · DASHBOARDS
Not started
Monthly performance against benchmark: Pipelines
Files flow from the sources through bronze and silver to gold, the tables dashboards read.
Span the builder: connecting and running
Pack version 1 · locked
The two Span roles talk it through here, so you see why the pack changed. People only approve the new version.
SPAN THE BUILDER · 11 OCT 2026, 22:18
I took design pack version 1 as it was approved. I connect each source as it says and run on a copy. I only come back to you when something fails.
Connecting the 4 sources
4 connected
administrator_positions
an uploaded file · as in the pack
2 files landed, latest administrator_positions_2026-09-30.xlsx.
Connected
global_equity_and_bond_i
an uploaded file · as in the pack
1 file landed, latest index_levels_2026-08_09.csv.
Connected
manager_statements
an uploaded file · as in the pack
Reading 3 marked fields from each statement (104 on file for the company).
Connected

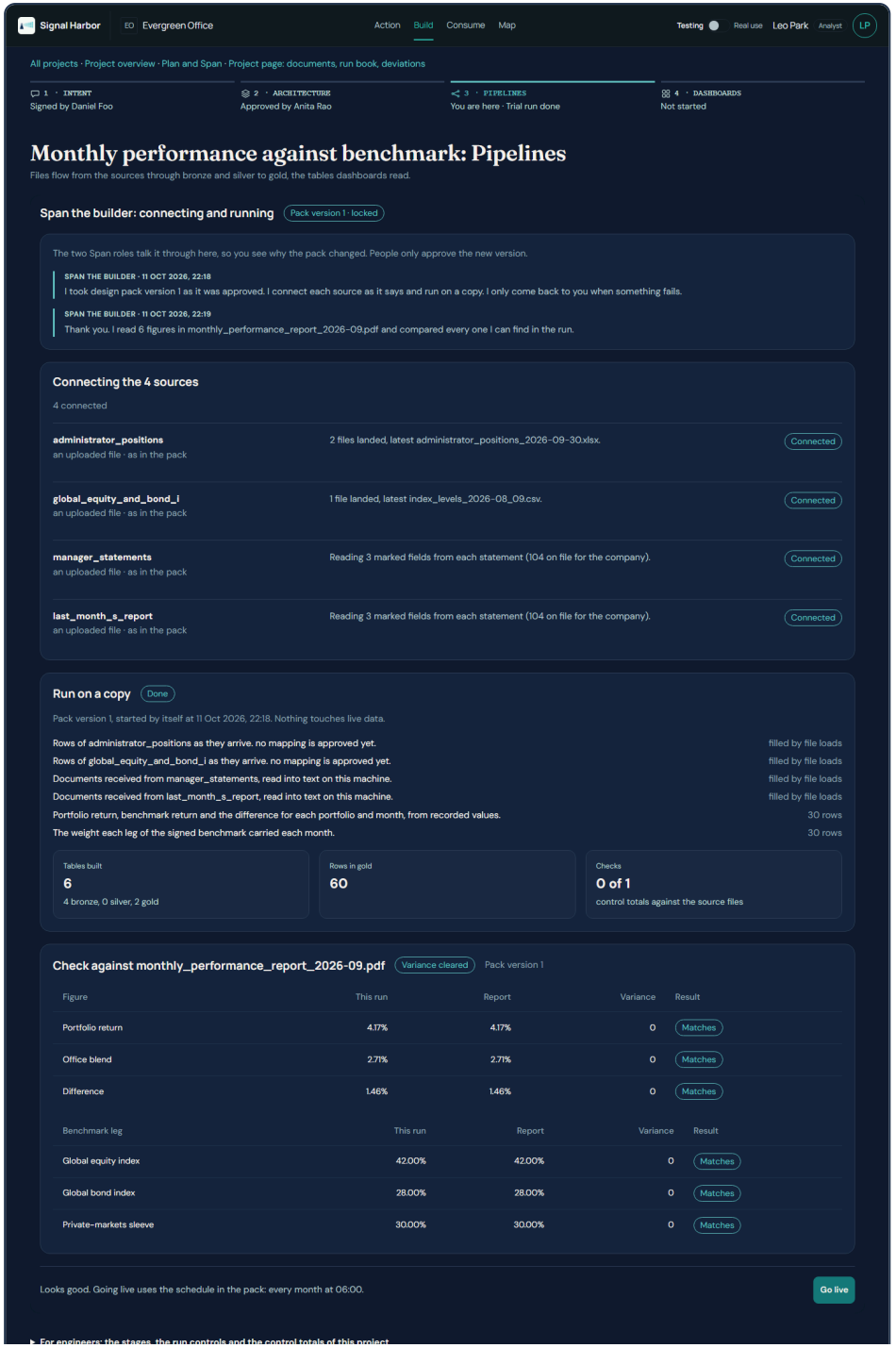

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You are here · Trial run done
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Not started
Monthly performance against benchmark: Pipelines
Files flow from the sources through bronze and silver to gold, the tables dashboards read.
Span the builder: connecting and running
Pack version 1 · locked
The two Span roles talk it through here, so you see why the pack changed. People only approve the new version.
SPAN THE BUILDER · 11 OCT 2026, 22:18
I took design pack version 1 as it was approved. I connect each source as it says and run on a copy. I only come back to you when something fails.
Connecting the 4 sources
4 connected
administrator_positions
an uploaded file · as in the pack
2 files landed, latest administrator_positions_2026-09-30.xlsx.
Connected
global_equity_and_bond_i
an uploaded file · as in the pack
1 file landed, latest index_levels_2026-08_09.csv.
Connected
manager_statements
an uploaded file · as in the pack
Reading 3 marked fields from each statement (104 on file for the company).
Connected
last_month_s_report
an uploaded file · as in the pack
Reading 3 marked fields from each statement (104 on file for the company).
Connected
Run on a copy
Done
Pack version 1, started by itself at 11 Oct 2026, 22:18. Nothing touches live data.
Rows of administrator_positions as they arrive. no mapping is approved yet.
filled by file loads
Rows of global_equity_and_bond_i as they arrive. no mapping is approved yet.
filled by file loads
Documents received from manager_statements, read into text on this machine.
filled by file loads
Documents received from last_month_s_report, read into text on this machine.
filled by file loads
Portfolio return, benchmark return and the difference for each portfolio and month, from recorded values.
30 rows
The weight each leg of the signed benchmark carried each month.
30 rows
Tables built
6
4 bronze, 0 silver, 2 gold
Rows in gold
60
Checks
0 of 1
control totals against the source files
SPAN THE BUILDER
Do you have last month's report to check against? I will compare every figure I can find in it.
Upload last month's report
Choose File No file chosen
Or pick a signed report
Choose one
Check against it
I have none: go live without a check
Give the builder a report to check against, then decide.
For engineers: the stages, the run controls and the control totals of this project

# 12 · Pipelines: benchmark request and variance

**Leo Park (data engineer), with Span the builder (agent)**

| | |
|---|---|
| **Intended** | The builder asks for last month's report, reads it and compares every figure it can find with the run, line by line. |
| **What we did** | Span asked: 'Do you have last month's report to check against? I will compare every figure I can find in it.' Leo uploaded the September report as issued (PDF). |
| **What happened** | 'Check against monthly_performance_report_2026-09.pdf: Variance cleared.' Portfolio return 4.17% against 4.17% (run 4.1667), Office blend 2.71% against 2.71% (run 2.7072), Difference 1.46% against 1.46% (run 1.4595), all Matches; legs: global equity 42%, global bond 28%, private-markets sleeve 30%, all Matches. 'Looks good. Going live uses the schedule in the pack: every month at 06:00.' There was no variance, so 'Send back to the architect' was not offered. |
| **Time taken** | 19 s |
| **If it goes wrong** | On an unexplained line Leo sends it back to Span the architect with the evidence; the architect issues the next pack version for Anita Rao to approve; the builder reruns by itself. |

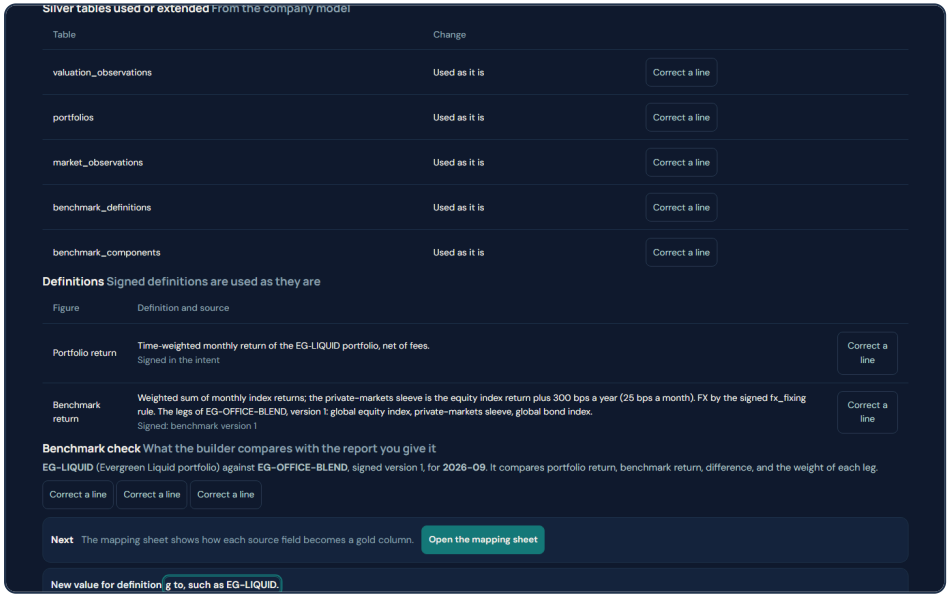

The weight each leg of the signed benchmark carried each month.
30 rows
Tables built
6
4 bronze, 0 silver, 2 gold
Rows in gold
60
Checks
0 of 1
control totals against the source files
Check against monthly_performance_report_2026-09.pdf
Variance cleared
Pack version 1
Figure
This run
Report
Variance
Result
Portfolio return
4.17%
4.17%
0
Matches
Office blend
2.71%
2.71%
0
Matches
Difference
1.46%
1.46%
0
Matches
Benchmark leg
This run
Report
Variance
Result
Global equity index
42.00%
42.00%
0
Matches
Global bond index
28.00%
28.00%
0
Matches
Private-markets sleeve
30.00%
30.00%
0
Matches
Looks good. Going live uses the schedule in the pack: every month at 06:00.
Go live
For engineers: the stages, the run controls and the control totals of this project

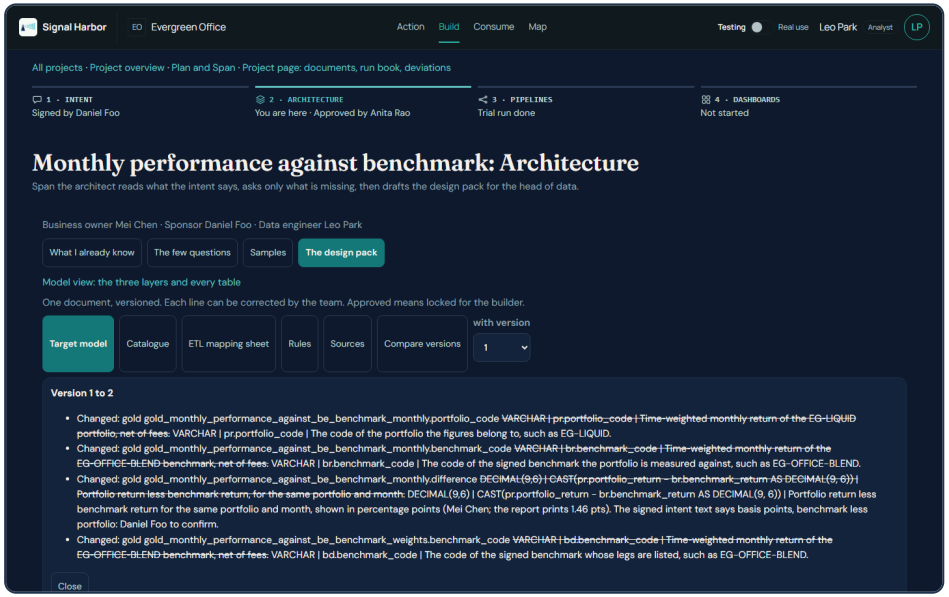

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1 · INTENT
Signed by Daniel Foo
2 · ARCHITECTURE
Approved by Anita Rao
3 · PIPELINES
You are here · Trial run done
4 · DASHBOARDS
Not started
Monthly performance against benchmark: Pipelines
Files flow from the sources through bronze and silver to gold, the tables dashboards read.
Span the builder: connecting and running
Pack version 1 · locked
The two Span roles talk it through here, so you see why the pack changed. People only approve the new version.
SPAN THE BUILDER · 11 OCT 2026, 22:18
I took design pack version 1 as it was approved. I connect each source as it says and run on a copy. I only come back to you when something fails.
SPAN THE BUILDER · 11 OCT 2026, 22:19
Thank you. I read 6 figures in monthly_performance_report_2026-09.pdf and compared every one I can find in the run.
Connecting the 4 sources
4 connected
administrator_positions
an uploaded file · as in the pack
2 files landed, latest administrator_positions_2026-09-30.xlsx.
Connected
global_equity_and_bond_i
an uploaded file · as in the pack
1 file landed, latest index_levels_2026-08_09.csv.
Connected
manager_statements
an uploaded file · as in the pack
Reading 3 marked fields from each statement (104 on file for the company).
Connected
last_month_s_report
an uploaded file · as in the pack
Reading 3 marked fields from each statement (104 on file for the company).
Connected
Run on a copy
Done
Pack version 1, started by itself at 11 Oct 2026, 22:18. Nothing touches live data.
Rows of administrator_positions as they arrive. no mapping is approved yet.
filled by file loads
Rows of global_equity_and_bond_i as they arrive. no mapping is approved yet.
filled by file loads
Documents received from manager_statements, read into text on this machine.
filled by file loads
Documents received from last_month_s_report, read into text on this machine.
filled by file loads
Portfolio return, benchmark return and the difference for each portfolio and month, from recorded values.
30 rows
The weight each leg of the signed benchmark carried each month.
30 rows
Tables built
6
4 bronze, 0 silver, 2 gold
Rows in gold
60
Checks
0 of 1
control totals against the source files
Check against monthly_performance_report_2026-09.pdf
Variance cleared
Pack version 1
Figure
This run
Report
Variance
Result
Portfolio return
4.17%
4.17%
0
Matches
Office blend
2.71%
2.71%
0
Matches
Difference
1.46%
1.46%
0
Matches
Benchmark leg
This run
Report
Variance
Result
Global equity index
42.00%
42.00%
0
Matches
Global bond index
28.00%
28.00%
0
Matches
Private-markets sleeve
30.00%
30.00%
0
Matches
Looks good. Going live uses the schedule in the pack: every month at 06:00.
Go live
For engineers: the stages, the run controls and the control totals of this project

# 13 · Pack version 2: Leo corrects the definitions

Leo Park (data engineer), as Span the architect's reviewer

| | |
|---|---|
| Intended | With no variance to send back, Leo reads the approved pack once more before going live and corrects what is wrong through a new version. |
| What we did | In the gold columns, Leo pressed Correct a line on four definitions: portfolio_code and benchmark_code (both tables) carried the text of a return KPI ('Time-weighted monthly return of the EG-LIQUID portfolio, net of fees'), and the difference column did not say its unit or that the signed intent text disagrees. He opened Compare versions (1 to 2) and sent version 2 to the head of data. |
| What happened | Correcting an approved pack forked version 2 as a draft (version 1 stays locked). Compare versions lists the four changes, old text struck through. Version 2: waiting for the head of data. The 'Send back to the architect' path was not exercised in this walk because the first run matched the report. |
| Time taken | 28 s |
| If it goes wrong | Anita Rao rejects version 2 with a note; version 1 stays the approved one. |



| | | |
|---|---|---|
| benchmark_components | Used as it is | Correct a line |

**Definitions** Signed definitions are used as they are

| Figure | Definition and source | |
|---|---|---|
| Portfolio return | Time-weighted monthly return of the EG-LIQUID portfolio, net of fees.<br>Signed in the intent | Correct a line |
| Benchmark return | Weighted sum of monthly index returns; the private-markets sleeve is the equity index return plus 300 bps a year (25 bps a month). FX by the signed fx_fixing rule. The legs of EG-OFFICE-BLEND, version 1: global equity index, private-markets sleeve, global bond index.<br>Signed: benchmark version 1 | Correct a line |

**Benchmark check** What the builder compares with the report you give it

**EG-LIQUID** (Evergreen Liquid portfolio) against **EG-OFFICE-BLEND**, signed version 1, for **2026-09**. It compares portfolio return, benchmark return, difference, and the weight of each leg.

Correct a line | Correct a line | Correct a line

**Next** The mapping sheet shows how each source field becomes a gold column. **Open the mapping sheet**

**Design pack version 2** Waiting for the head of data

4 sources · 2 gold and 5 silver tables · 19 mapping rows (0 open) · 1 checks

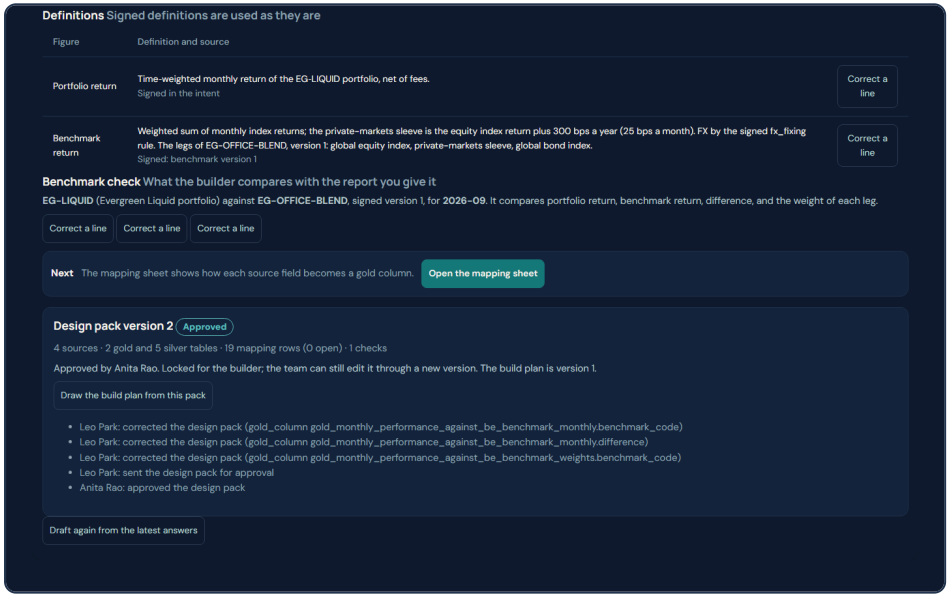
- Leo Park: corrected the design pack (gold_column gold_monthly_performance_against_be_benchmark_monthly.portfolio_code)
- Leo Park: corrected the design pack (gold_column gold_monthly_performance_against_be_benchmark_monthly.benchmark_code)
- Leo Park: corrected the design pack (gold_column gold_monthly_performance_against_be_benchmark_monthly.difference)
- Leo Park: corrected the design pack (gold_column gold_monthly_performance_against_be_benchmark_weights.benchmark_code)
- Leo Park: sent the design pack for approval

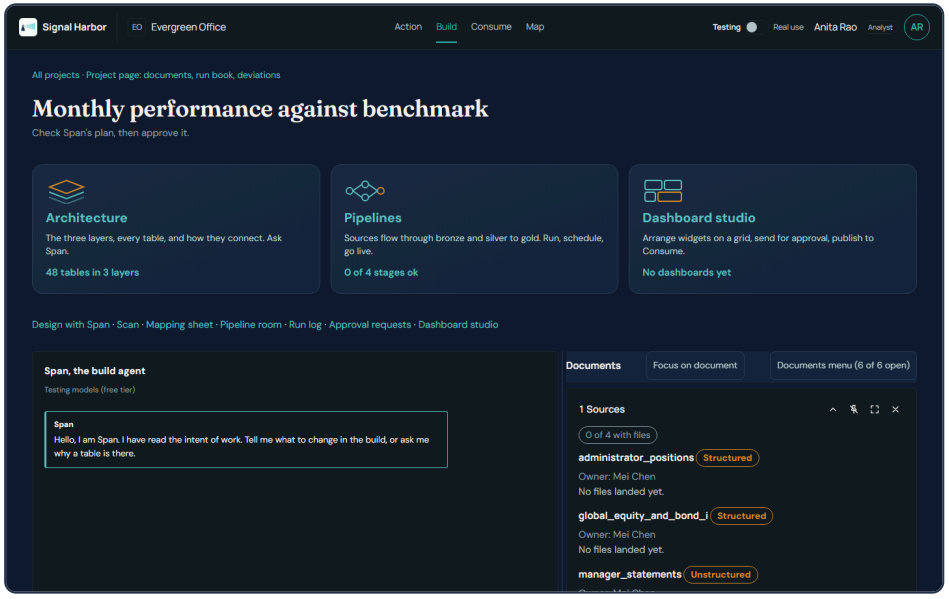
Draft again from the latest answers

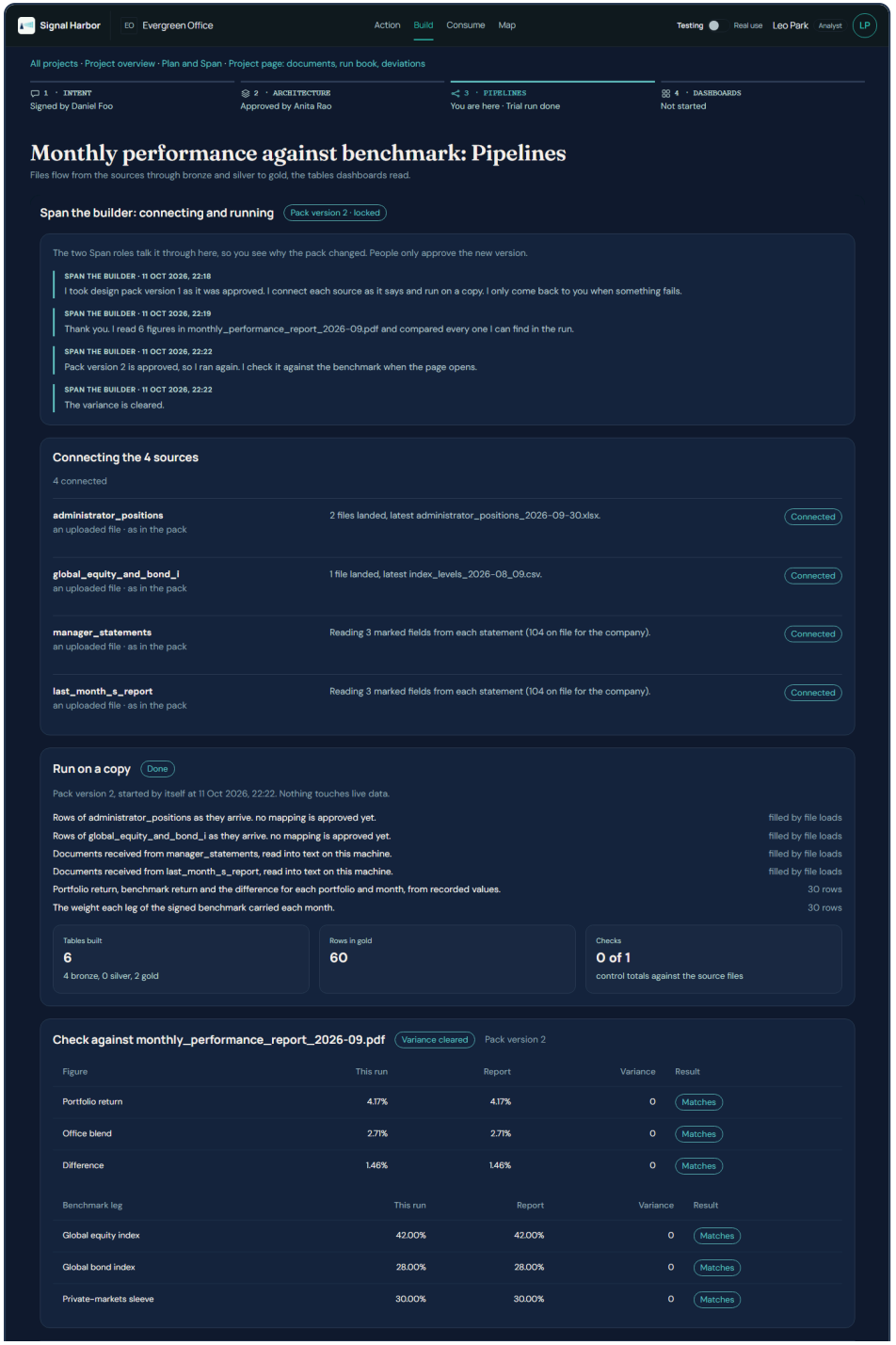
# 14 · Approve pack version 2; the builder reruns by itself

**Anita Rao (head of data), Leo Park (data engineer), with Span the builder (agent)**

| | |
|---|---|
| Intended | The head of data approves the new version; the builder reruns on a copy without being asked and checks the report again. |
| What we did | Anita pressed Approve version 2; Leo drew plan version 2 from it; Anita approved the plan ('Plan version 2 from pack version 2 (definitions only)'). Leo reopened Pipelines. |
| What happened | Pack version 2 approved and locked. The builder ran pack version 2 on a copy by itself and checked it against the same report: 'Variance cleared', Pack version 2, the three figures and three legs all Matches. |
| Time taken | 59 s |
| If it goes wrong | If the rerun fails or a line no longer matches, the builder holds it and Leo sends it back to the architect; Anita approves the fix. |




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Signed by Daniel Foo
2 · ARCHITECTURE
Approved by Anita Rao
3 · PIPELINES
You are here · Trial run done
4 · DASHBOARDS
Not started
Monthly performance against benchmark: Pipelines
Files flow from the sources through bronze and silver to gold, the tables dashboards read.
Span the builder: connecting and running
Pack version 2 · locked
The two Span roles talk it through here, so you see why the pack changed. People only approve the new version.
SPAN THE BUILDER · 11 OCT 2026, 22:18
I took design pack version 1 as it was approved. I connect each source as it says and run on a copy. I only come back to you when something fails.
SPAN THE BUILDER · 11 OCT 2026, 22:19
Thank you. I read 6 figures in monthly_performance_report_2026-09.pdf and compared every one I can find in the run.
SPAN THE BUILDER · 11 OCT 2026, 22:22
Pack version 2 is approved, so I ran again. I check it against the benchmark when the page opens.
SPAN THE BUILDER · 11 OCT 2026, 22:22
The variance is cleared.
Connecting the 4 sources
4 connected
administrator_positions
an uploaded file · as in the pack
2 files landed, latest administrator_positions_2026-09-30.xlsx.
Connected
global_equity_and_bond_i
an uploaded file · as in the pack
1 file landed, latest index_levels_2026-08_09.csv.
Connected
manager_statements
an uploaded file · as in the pack
Reading 3 marked fields from each statement (104 on file for the company).
Connected
last_month_s_report
an uploaded file · as in the pack
Reading 3 marked fields from each statement (104 on file for the company).
Connected
Run on a copy
Done
Pack version 2, started by itself at 11 Oct 2026, 22:22. Nothing touches live data.
Rows of administrator_positions as they arrive. no mapping is approved yet.
filled by file loads
Rows of global_equity_and_bond_i as they arrive. no mapping is approved yet.
filled by file loads
Documents received from manager_statements, read into text on this machine.
filled by file loads
Documents received from last_month_s_report, read into text on this machine.
filled by file loads
Portfolio return, benchmark return and the difference for each portfolio and month, from recorded values.
30 rows
The weight each leg of the signed benchmark carried each month.
30 rows
Tables built
6
4 bronze, 0 silver, 2 gold
Rows in gold
60
Checks
0 of 1
control totals against the source files
Check against monthly_performance_report_2026-09.pdf
Variance cleared
Pack version 2
Figure
This run
Report
Variance
Result
Portfolio return
4.17%
4.17%
0
Matches
Office blend
2.71%
2.71%
0
Matches
Difference
1.46%
1.46%
0
Matches
Benchmark leg
This run
Report
Variance
Result
Global equity index
42.00%
42.00%
0
Matches
Global bond index
28.00%
28.00%
0
Matches
Private-markets sleeve
30.00%
30.00%
0
Matches




# 15 · Go live

**Leo Park (data engineer), with Span the builder (agent)**

| | |
|---|---|
| Intended | With the check clean, take the pipeline live on the pack's schedule. |
| What we did | Leo pressed Go live on the Pipelines step. |
| What happened | The round is live: 'Live on the pack schedule, every month at 06:00. Lumen proposes a starter dashboard.' The builder hands on to Lumen for a starter dashboard. |
| Time taken | 18 s |
| If it goes wrong | Leo or Anita Rao stops the schedule; a failed scheduled run comes back to Leo in Action. |

Files flow from the sources through bronze and silver to gold, the tables dashboards read.

## Span the builder: connecting and running

Pack version 2 · locked

The two Span roles talk it through here, so you see why the pack changed. People only approve the new version.

> **SPAN THE BUILDER · 11 OCT 2026, 22:18**
> I took design pack version 1 as it was approved. I connect each source as it says and run on a copy. I only come back to you when something fails.

> **SPAN THE BUILDER · 11 OCT 2026, 22:19**
> Thank you. I read 6 figures in monthly_performance_report_2026-09.pdf and compared every one I can find in the run.

> **SPAN THE BUILDER · 11 OCT 2026, 22:22**
> Pack version 2 is approved, so I ran again. I check it against the benchmark when the page opens.

> **SPAN THE BUILDER · 11 OCT 2026, 22:22**
> The variance is cleared.

> **SPAN THE BUILDER · 11 OCT 2026, 22:23**
> Leo Park took pack version 2 live on the pack schedule (monthly at 06:00). Lumen proposes a starter dashboard next.

## Connecting the 4 sources

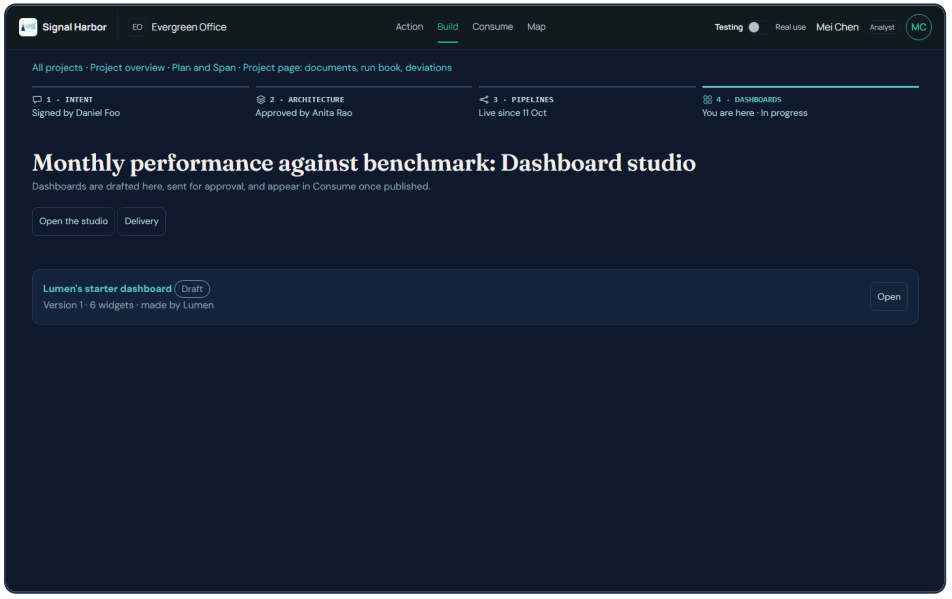
4 connected

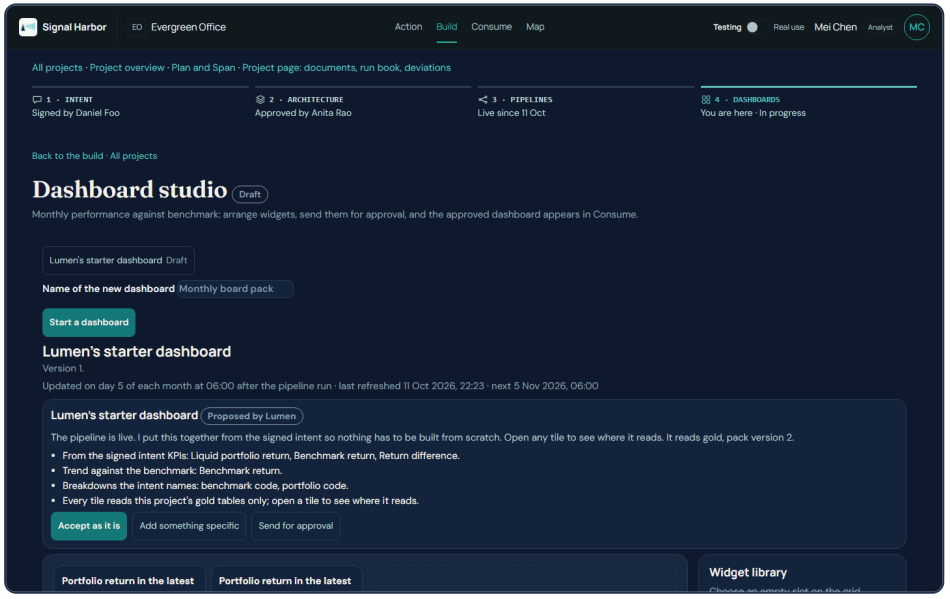
| | | |
|---|---|---|
| **administrator_positions**<br>an uploaded file · as in the pack | 2 files landed, latest administrator_positions_2026-09-30.xlsx. | Connected |
| **global_equity_and_bond_i**<br>an uploaded file · as in the pack | 1 file landed, latest index_levels_2026-08_09.csv. | Connected |
| **manager_statements**<br>an uploaded file · as in the pack | Reading 3 marked fields from each statement (104 on file for the company). | Connected |

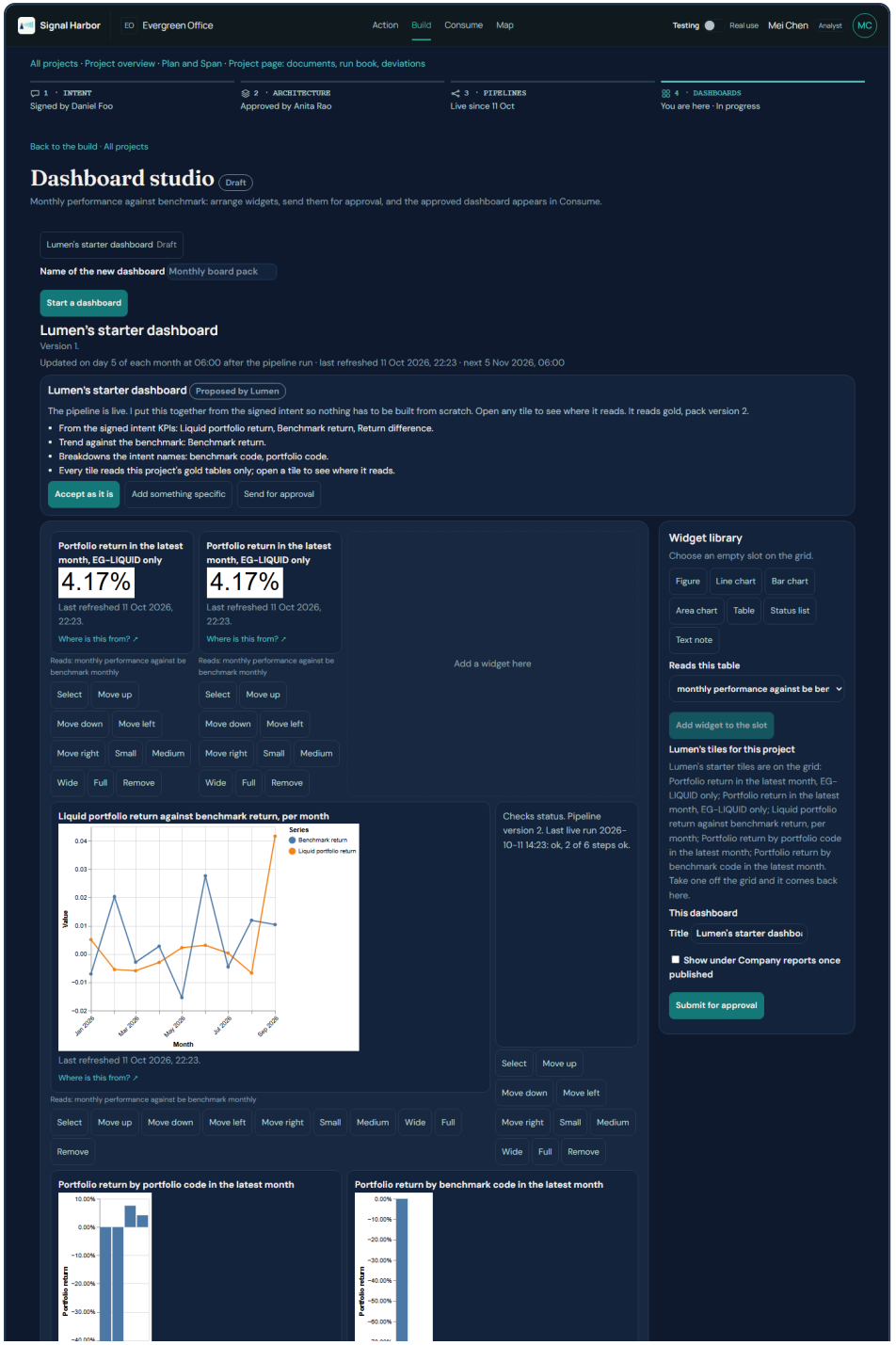
# 16 · Lumen's starter dashboard

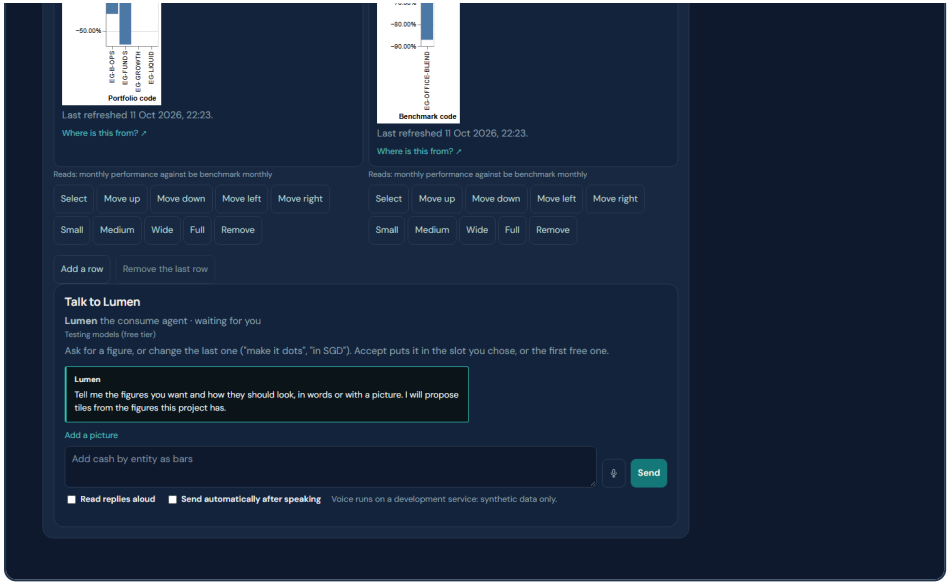
Mei Chen (business owner), with Lumen (agent)

| | |
|---|---|
| Intended | When the pipeline goes live, Lumen proposes a starter dashboard from the intent's figures, readers and breakdowns, so nothing is built from scratch. |
| What we did | Mei opened the Dashboards step: 'Lumen's starter dashboard, Draft, version 1, 6 widgets, made by Lumen', and opened it in the studio. |
| What happened | Lumen: 'The pipeline is live. I put this together from the signed intent so nothing has to be built from scratch... It reads gold, pack version 2.' Tiles: Portfolio return in the latest month, EG-LIQUID only, 4.17% (twice, the same tile); a monthly trend of liquid portfolio return against benchmark return (axis in fractions, -0.02 to 0.04); portfolio return by portfolio code (all four portfolios, bars down to about -50%); portfolio return by benchmark code (one bar near -90%); a checks status line ('2 of 6 steps ok'). No benchmark return or difference figure tile, although the intent names both. The refresh line reads 'day 5 of each month at 06:00; next 5 Nov 2026'. |
| Time taken | 20 s |
| If it goes wrong | Mei removes or asks Lumen to change a tile; Leo checks a figure that does not match gold ('Where is this from?'). |



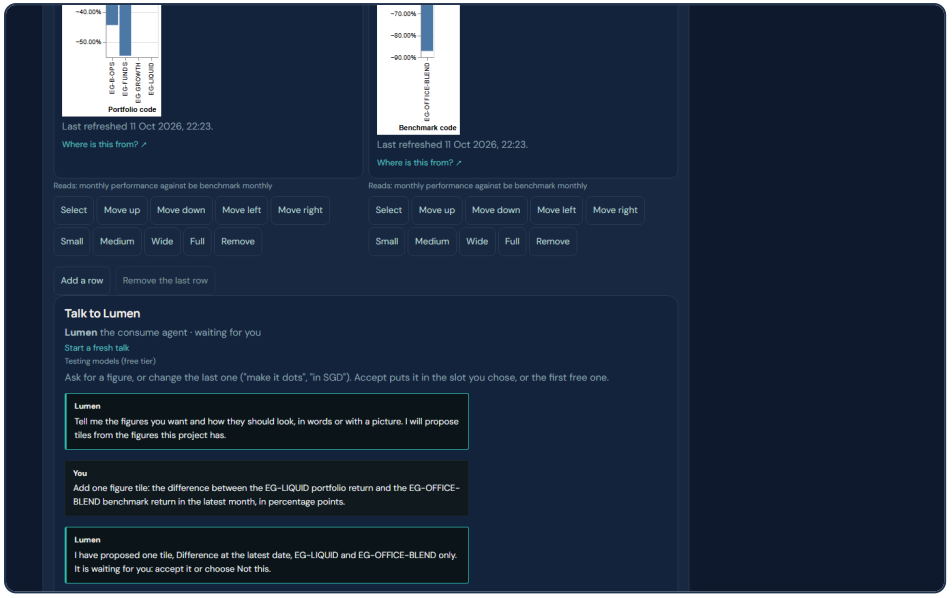

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1 · INTENT
Signed by Daniel Foo
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Approved by Anita Rao
3 · PIPELINES
Live since 11 Oct
4 · DASHBOARDS
You are here · In progress
Back to the build · All projects
Dashboard studio
Draft
Monthly performance against benchmark: arrange widgets, send them for approval, and the approved dashboard appears in Consume.
Lumen's starter dashboard Draft
Name of the new dashboard Monthly board pack
Start a dashboard
Lumen's starter dashboard
Version 1.
Updated on day 5 of each month at 06:00 after the pipeline run · last refreshed 11 Oct 2026, 22:23 · next 5 Nov 2026, 06:00
Lumen's starter dashboard
Proposed by Lumen
The pipeline is live. I put this together from the signed intent so nothing has to be built from scratch. Open any tile to see where it reads. It reads gold, pack version 2.
From the signed intent KPIs: Liquid portfolio return, Benchmark return, Return difference.
Trend against the benchmark: Benchmark return.
Breakdowns the intent names: benchmark code, portfolio code.
Every tile reads this project's gold tables only; open a tile to see where it reads.
Accept as it is
Add something specific
Send for approval
Portfolio return in the latest month, EG-LIQUID only
4.17%
Last refreshed 11 Oct 2026, 22:23.
Where is this from? ↗
Reads: monthly performance against be benchmark monthly
Select
Move up
Move down
Move left
Move right
Small
Medium
Wide
Full
Remove
Portfolio return in the latest month, EG-LIQUID only
4.17%
Last refreshed 11 Oct 2026, 22:23.
Where is this from? ↗
Reads: monthly performance against be benchmark monthly
Add a widget here
Liquid portfolio return against benchmark return, per month
Series
Benchmark return
Liquid portfolio return
Value
Month
Last refreshed 11 Oct 2026, 22:23.
Where is this from? ↗
Reads: monthly performance against be benchmark monthly
Checks status. Pipeline version 2. Last live run 2026-10-11 14:23: ok, 2 of 6 steps ok.
Widget library
Choose an empty slot on the grid.
Figure
Line chart
Bar chart
Area chart
Table
Status list
Text note
Reads this table
monthly performance against be ber
Add widget to the slot
Lumen's tiles for this project
Lumen's starter tiles are on the grid: Portfolio return in the latest month, EG-LIQUID only; Portfolio return in the latest month, EG-LIQUID only; Liquid portfolio return against benchmark return, per month; Portfolio return by portfolio code in the latest month; Portfolio return by benchmark code in the latest month. Take one off the grid and it comes back here.
This dashboard
Title Lumen's starter dashbo
Show under Company reports once published
Submit for approval
Portfolio return by portfolio code in the latest month
Portfolio return
Portfolio return by benchmark code in the latest month
Portfolio return

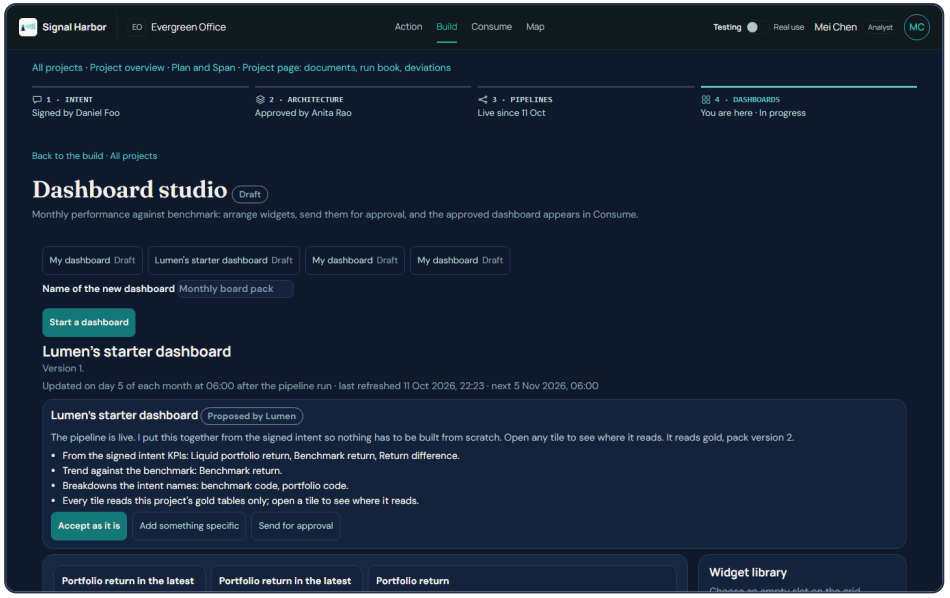


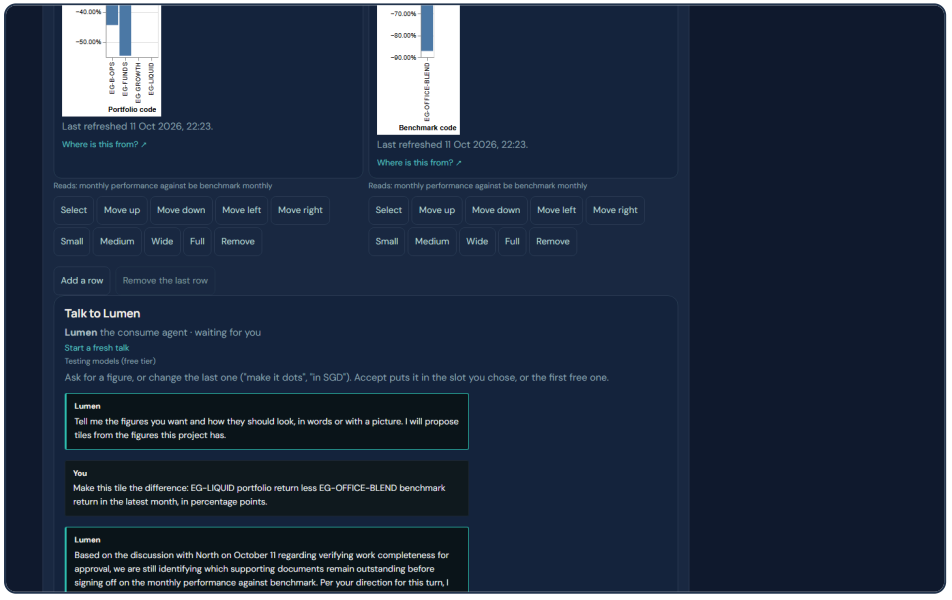

# 17 · Mei adds one specific tile

## Mei Chen (business owner), with Lumen (agent)

| | |
|---|---|
| Intended | Mei asks Lumen for one specific addition, the difference in percentage points for the latest month, and accepts it onto the starter. |
| What we did | Mei asked Lumen in the studio's Talk to Lumen: 'Add one figure tile: the difference between the EG-LIQUID portfolio return and the EG-OFFICE-BLEND benchmark return in the latest month, in percentage points.' (twice, once with the starter opened by its own link). Then she used the widget library (empty slot, Figure, table '...benchmark monthly', Add widget to the slot). Then she tried to select that tile and ask Lumen to turn it into the difference. |
| What happened | Lumen replied 'I have proposed one tile, Difference at the latest date, EG-LIQUID and EG-OFFICE-BLEND only. It is waiting for you: accept it or choose Not this.', but no Accept button appeared: each ask placed the proposal on a new, empty personal 'My dashboard' (three were made), not on the starter the studio shows. The widget library added a figure 'Portfolio return' (the first column), not the difference, and offers no column choice. Selecting the tile did not register, and Lumen's next reply went off topic ('...I am holding off on the chart request to focus solely on the approval workflow'). The requested difference tile was NOT added; the starter carries one extra 'Portfolio return' figure. |
| Time taken | 4 min 47 s |
| If it goes wrong | Leo Park adds the tile in the studio from the gold column; the Lumen defect goes to the product team. |


−40.00%
−50.00%
EG-B-OPS
EG-I-MKS
EG-D-MKT
EG-LIQUID
Portfolio code
Last refreshed 11 Oct 2026, 22:23.
Where is this from? ↗
−70.00%
−80.00%
−90.00%
EG-OFFICE-BLEND
Benchmark code
Last refreshed 11 Oct 2026, 22:23.
Where is this from? ↗
Reads: monthly performance against be benchmark monthly
Select
Move up
Move down
Move left
Move right
Small
Medium
Wide
Full
Remove
Add a row
Remove the last row
Talk to Lumen
Lumen the consume agent · waiting for you
Start a fresh talk
Testing models (free tier)
Ask for a figure, or change the last one ("make it dots", "in SGD"). Accept puts it in the slot you chose, or the first free one.
Lumen
Tell me the figures you want and how they should look, in words or with a picture. I will propose tiles from the figures this project has.
You
Add one figure tile: the difference between the EG-LIQUID portfolio return and the EG-OFFICE-BLEND benchmark return in the latest month, in percentage points.
Lumen
I have proposed one tile, Difference at the latest date, EG-LIQUID and EG-OFFICE-BLEND only. It is waiting for you: accept it or choose Not this.

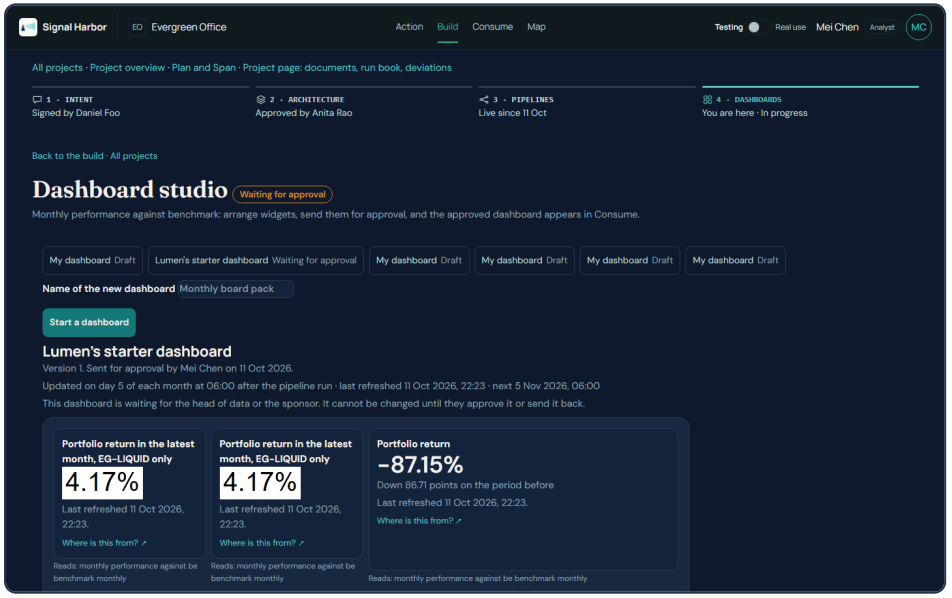

Signal Harbor
EO Evergreen Office
Action
Build
Consume
Map
Testing
Real use
Mei Chen
Analyst
MC
All projects · Project overview · Plan and Span · Project page: documents, run book, deviations
1 · INTENT
Signed by Daniel Foo
2 · ARCHITECTURE
Approved by Anita Rao
3 · PIPELINES
Live since 11 Oct
4 · DASHBOARDS
You are here · In progress
Back to the build · All projects
Dashboard studio
Draft
Monthly performance against benchmark: arrange widgets, send them for approval, and the approved dashboard appears in Consume.
My dashboard Draft
Lumen's starter dashboard Draft
My dashboard Draft
My dashboard Draft
Name of the new dashboard
Monthly board pack
Start a dashboard
Lumen's starter dashboard
Version 1.
Updated on day 5 of each month at 06:00 after the pipeline run · last refreshed 11 Oct 2026, 22:23 · next 5 Nov 2026, 06:00
Lumen's starter dashboard
Proposed by Lumen
The pipeline is live. I put this together from the signed intent so nothing has to be built from scratch. Open any tile to see where it reads. It reads gold, pack version 2.
From the signed intent KPIs: Liquid portfolio return, Benchmark return, Return difference.
Trend against the benchmark: Benchmark return.
Breakdowns the intent names: benchmark code, portfolio code.
Every tile reads this project's gold tables only; open a tile to see where it reads.
Accept as it is
Add something specific
Send for approval
Portfolio return in the latest
Portfolio return in the latest
Portfolio return
Widget library

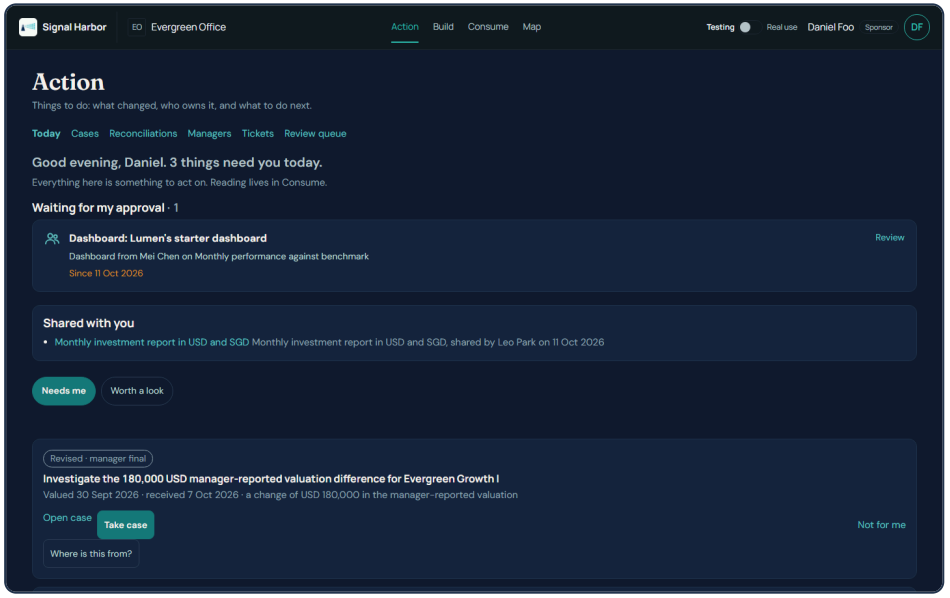

−40.00%
−50.00%
EG-B-OPS
EG-I-MKS
EG-D-MKT
EG-LIQUID
Portfolio code
Last refreshed 11 Oct 2026, 22:23.
Where is this from? ↗
−70.00%
−80.00%
−90.00%
EG-OFFICE-BLEND
Benchmark code
Last refreshed 11 Oct 2026, 22:23.
Where is this from? ↗
Reads: monthly performance against be benchmark monthly
Select
Move up
Move down
Move left
Move right
Small
Medium
Wide
Full
Remove
Add a row
Remove the last row
Talk to Lumen
Lumen the consume agent · waiting for you
Start a fresh talk
Testing models (free tier)
Ask for a figure, or change the last one ("make it dots", "in SGD"). Accept puts it in the slot you chose, or the first free one.
Lumen
Tell me the figures you want and how they should look, in words or with a picture. I will propose tiles from the figures this project has.
You
Make this tile the difference: EG-LIQUID portfolio return less EG-OFFICE-BLEND benchmark return in the latest month, in percentage points.
Lumen
Based on the discussion with North on October 11 regarding verifying work completeness for approval, we are still identifying which supporting documents remain outstanding before signing off on the monthly performance against benchmark. Per your direction for this turn, I

# 18 · Daniel approves the dashboard

Mei Chen, then Daniel Foo (sponsor)

| | |
|---|---|
| Intended | Mei sends the dashboard for approval; someone other than its maker approves it. |
| What we did | Mei pressed Send for approval on the starter. Daniel opened Action, then the dashboard, read 'What Lumen checked' and pressed Approve. |
| What happened | Daniel's Action showed 'Dashboard: Lumen's starter dashboard, from Mei Chen'. 'What Lumen checked': tiles name their source (checked); readers are the project's people (checked); but 'The month's numbers - not checked: No reviewed run is on record for this project' and 'The benchmark is the signed definition - not checked: No signed benchmark version is on record', although the builder's check matched the report and EG-OFFICE-BLEND version 1 is signed. Daniel approved; the dashboard is published. |
| Time taken | 45 s |
| If it goes wrong | Daniel sends it back with a reason; Mei changes the tiles and sends it again. |



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All projects · Project overview · Plan and Span · Project page: documents, run book, deviations

1 · INTENT
Signed by Daniel Foo

2 · ARCHITECTURE
Approved by Anita Rao

3 · PIPELINES
Live since 11 Oct

4 · DASHBOARDS
You are here · Waiting for Daniel Foo's approval

Back to the build · All projects

# Dashboard studio

Waiting for approval

Monthly performance against benchmark: arrange widgets, send them for approval, and the approved dashboard appears in Consume.

Lumen's starter dashboard Waiting for approval, needs you

## Lumen's starter dashboard

Version 1. Sent for approval by Mei Chen on 11 Oct 2026.

Updated on day 5 of each month at 06:00 after the pipeline run · last refreshed 11 Oct 2026, 22:23 · next 5 Nov 2026, 06:00

WAITING FOR YOUR APPROVAL

### Approve the dashboard before it goes to the council

Before this: Mei Chen sent it for approval on 11 Oct 2026 at 22:39.

Check the preview below, which uses the live figures, and the sources of each widget.

- Portfolio return in the latest month, EG-LIQUID only: reads monthly performance against be benchmark monthly
- Portfolio return in the latest month, EG-LIQUID only: reads monthly performance against be benchmark monthly
- Liquid portfolio return against benchmark return, per month: reads monthly performance against be benchmark monthly
- Portfolio return by portfolio code in the latest month: reads monthly performance against be benchmark monthly
- Portfolio return by benchmark code in the latest month: reads monthly performance against be benchmark monthly
- Checks status: a text note
- Portfolio return: reads monthly performance against be benchmark monthly

**What Lumen checked**

– **The month's numbers** · not checked
No reviewed run is on record for this project, so Lumen could not compare them.

Signal Harbor | EO Evergreen Office | Action | Build | Consume | Map | Testing | Real use | Daniel Foo | Sponsor | DF

All projects · Project overview · Plan and Span · Project page: documents, run book, deviations

1 · INTENT
Signed by Daniel Foo

2 · ARCHITECTURE
Approved by Anita Rao

3 · PIPELINES
Live since 11 Oct

4 · DASHBOARDS
You are here · Waiting for Daniel Foo's approval

Back to the build · All projects

# Dashboard studio

Published

Monthly performance against benchmark: arrange widgets, send them for approval, and the approved dashboard appears in Consume.

Lumen's starter dashboard Published

## Lumen's starter dashboard

Version 1. Approved by Daniel Foo on 11 Oct 2026.

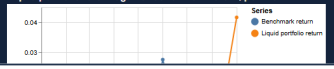
Updated on day 5 of each month at 06:00 after the pipeline run · last refreshed 11 Oct 2026, 22:23 · next 5 Nov 2026, 06:00

**Portfolio return in the latest month, EG-LIQUID only**
4.17%
Last refreshed 11 Oct 2026, 22:23.
Where is this from? ↗
Reads: monthly performance against be benchmark monthly

**Portfolio return in the latest month, EG-LIQUID only**
4.17%
Last refreshed 11 Oct 2026, 22:23.
Where is this from? ↗
Reads: monthly performance against be benchmark monthly

**Portfolio return**
−87.15%
Down 86.71 points on the period before
Last refreshed 11 Oct 2026, 22:23.
Where is this from? ↗
Reads: monthly performance against be benchmark monthly

**Liquid portfolio return against benchmark return, per month**

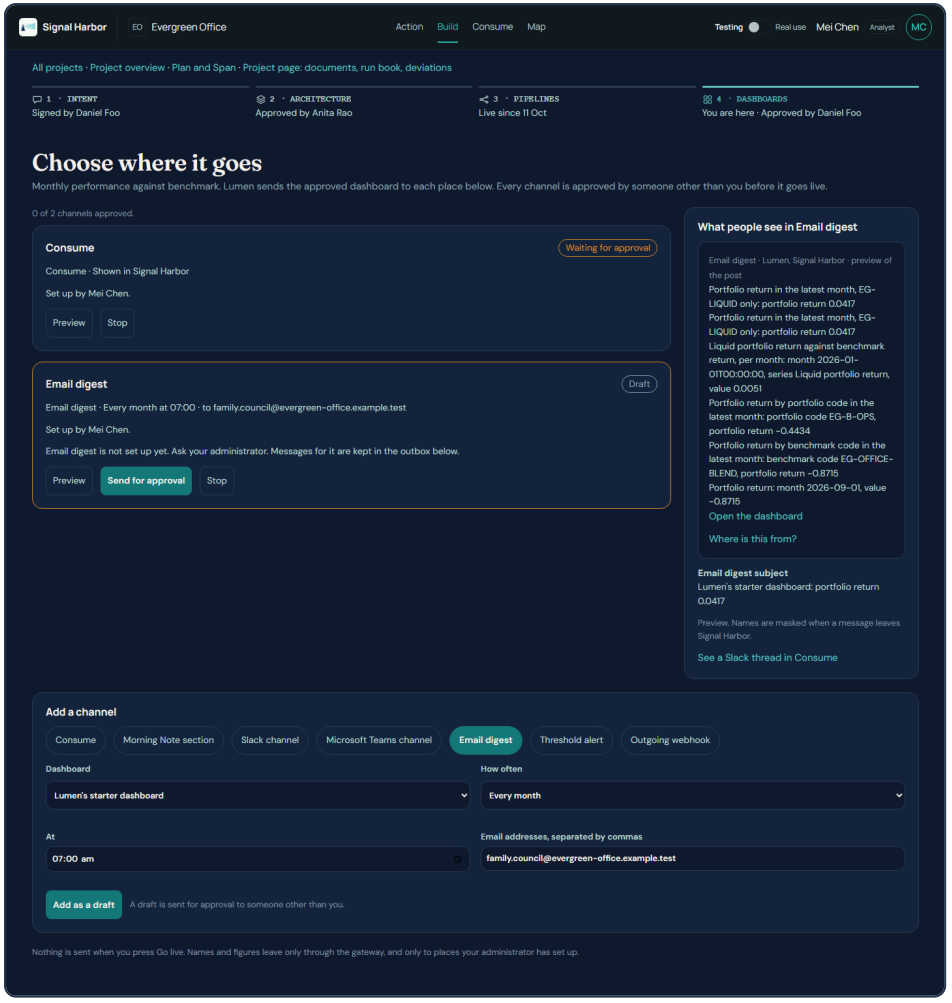


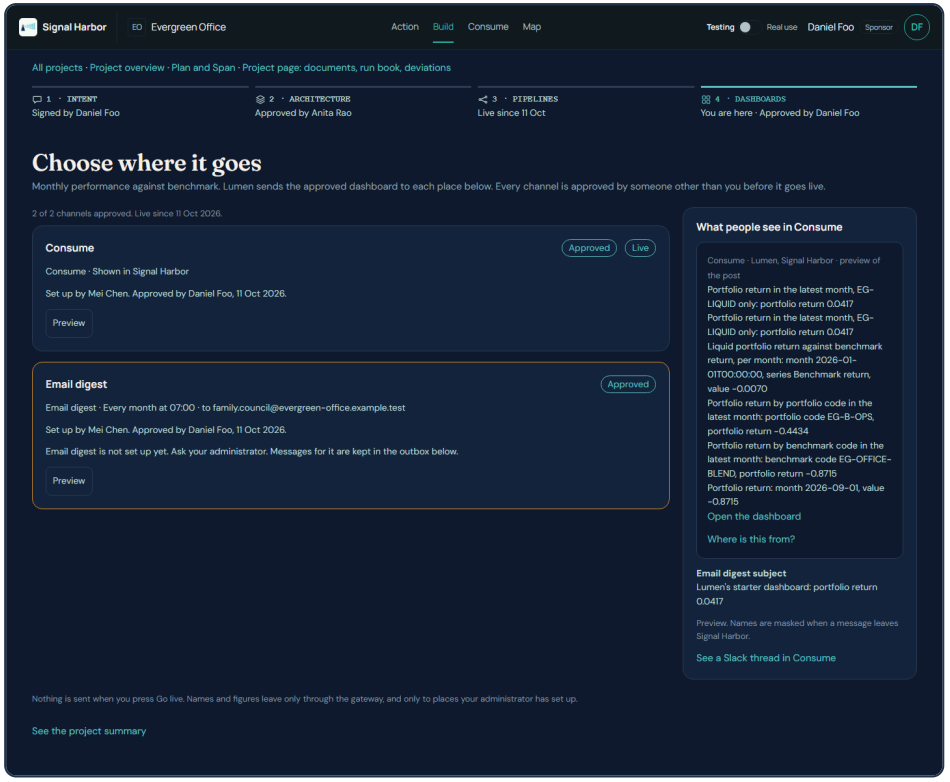

Checks status. Pipeline version 2. Last live run 2026-10-11 14:23: ok, 2 of 6 steps ok.

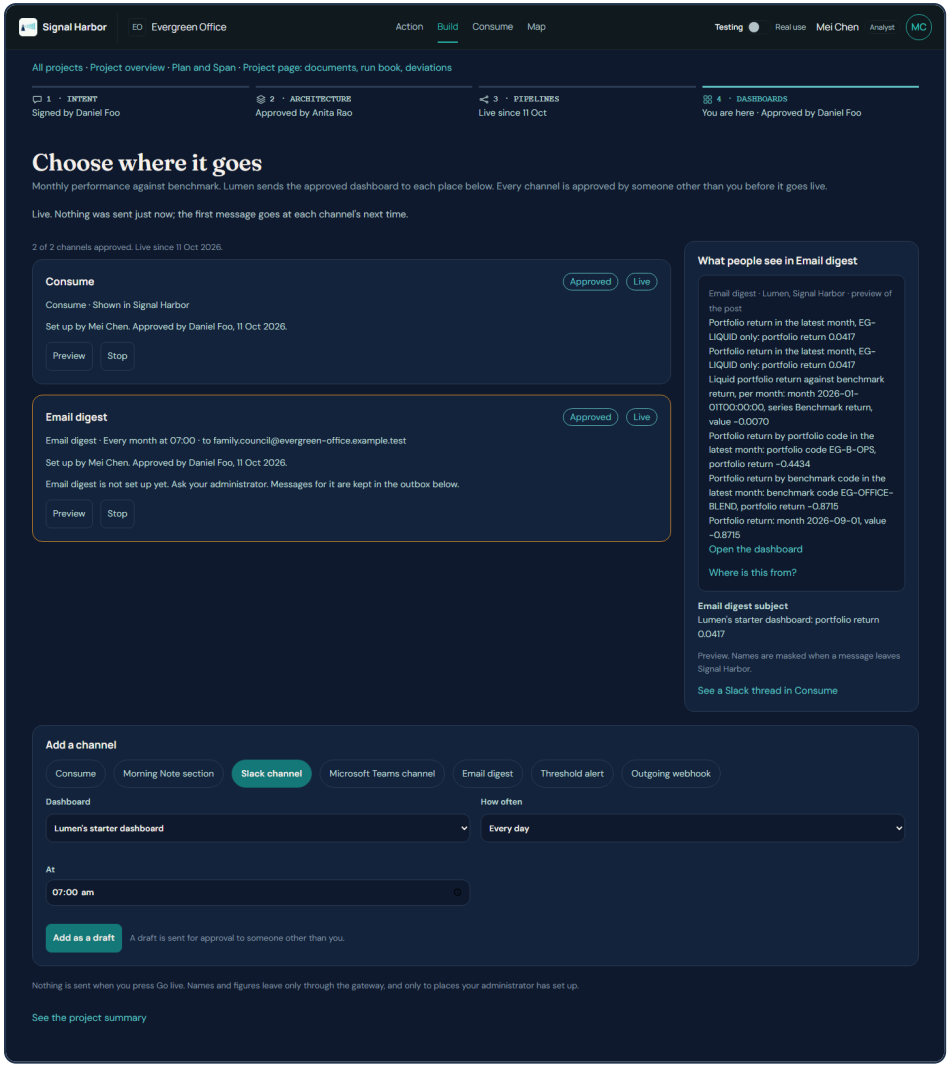
# 19 · Delivery: Consume and the council email

**Mei Chen (business owner), Daniel Foo (sponsor)**

| | |
|---|---|
| Intended | Mei chooses where the approved dashboard goes (Consume and a monthly email digest to the council); someone else approves each channel; then delivery goes live. |
| What we did | Mei added Consume and an Email digest (monthly, 07:00, to family.council@evergreen-office.example.test, a synthetic address), opened the email Preview and sent both for approval; Daniel approved both; Mei pressed Go live. |
| What happened | Both channels approved and live. The email preview (subject 'Lumen's starter dashboard: portfolio return 0.0417') lists each tile in words, e.g. 'Portfolio return in the latest month, EG-LIQUID only: portfolio return 0.0417' and 'Portfolio return by benchmark code in the latest month: benchmark code EG-OFFICE-BLEND, portfolio return -0.8715'. Figures are raw fractions, not percentages, and the implausible bar values travel into the email. The outbox is empty: 'Nothing is sent when you press Go live'; the first message goes at the next scheduled time, so no sent email could be shown. The first attempt left the email channel in draft (only Consume was submitted); it was submitted and approved on a second pass. |
| Time taken | 2 min 01 s |
| If it goes wrong | Daniel sends a channel back with a reason, or Mei stops it; a failed send stays in the outbox with Send again. |


Signal Harbor
EO Evergreen Office
Action Build Consume Map
Testing Real use Mei Chen Analyst MC
All projects · Project overview · Plan and Span · Project page: documents, run book, deviations
1 · INTENT
Signed by Daniel Foo
2 · ARCHITECTURE
Approved by Anita Rao
3 · PIPELINES
Live since 11 Oct
4 · DASHBOARDS
You are here · Approved by Daniel Foo
Choose where it goes
Monthly performance against benchmark. Lumen sends the approved dashboard to each place below. Every channel is approved by someone other than you before it goes live.
0 of 2 channels approved.
Consume
Waiting for approval
Consume · Shown in Signal Harbor
Set up by Mei Chen.
Preview
Stop
Email digest
Draft
Email digest · Every month at 07:00 · to family.council@evergreen-office.example.test
Set up by Mei Chen.
Email digest is not set up yet. Ask your administrator. Messages for it are kept in the outbox below.
Preview
Send for approval
Stop
What people see in Email digest
Email digest · Lumen, Signal Harbor · preview of the post
Portfolio return in the latest month, EG-LIQUID only: portfolio return 0.0417
Portfolio return in the latest month, EG-LIQUID only: portfolio return 0.0417
Liquid portfolio return against benchmark return, per month: month 2026-01-01T00:00:00, series Liquid portfolio return, value 0.0051
Portfolio return by portfolio code in the latest month: portfolio code EG-B-OPS, portfolio return -0.4434
Portfolio return by benchmark code in the latest month: benchmark code EG-OFFICE-BLEND, portfolio return -0.8715
Portfolio return: month 2026-09-01, value -0.8715
Open the dashboard
Where is this from?
Email digest subject
Lumen's starter dashboard: portfolio return 0.0417
Preview. Names are masked when a message leaves Signal Harbor.
See a Slack thread in Consume
Add a channel
Consume
Morning Note section
Slack channel
Microsoft Teams channel
Email digest
Threshold alert
Outgoing webhook
Dashboard
Lumen's starter dashboard
How often
Every month
At
07:00 am
Email addresses, separated by commas
family.council@evergreen-office.example.test
Add as a draft
A draft is sent for approval to someone other than you.
Nothing is sent when you press Go live. Names and figures leave only through the gateway, and only to places your administrator has set up.

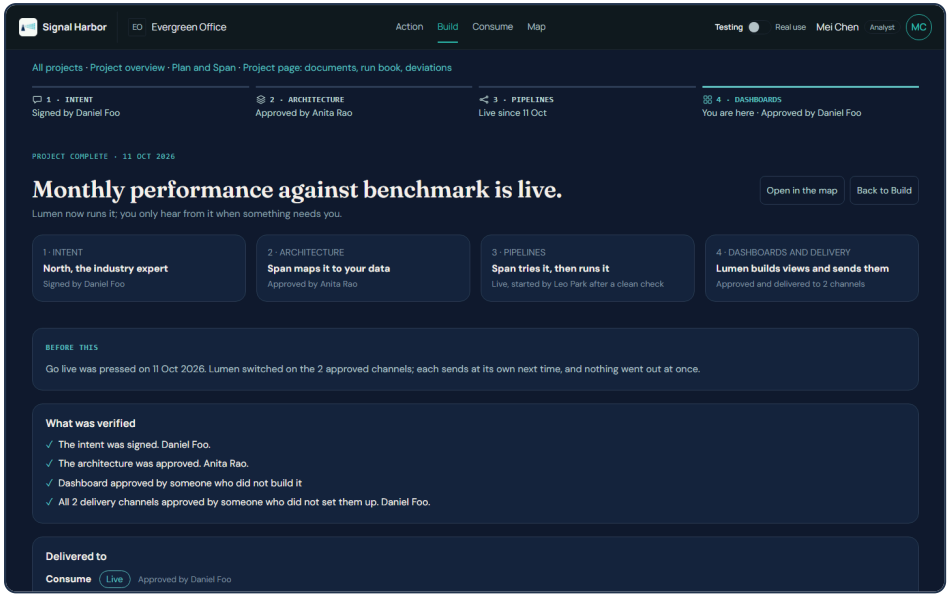

Signal Harbor
EO Evergreen Office
Action Build Consume Map
Testing Real use Daniel Foo Sponsor DF
All projects · Project overview · Plan and Span · Project page: documents, run book, deviations
1 · INTENT
Signed by Daniel Foo
2 · ARCHITECTURE
Approved by Anita Rao
3 · PIPELINES
Live since 11 Oct
4 · DASHBOARDS
You are here · Approved by Daniel Foo
Choose where it goes
Monthly performance against benchmark. Lumen sends the approved dashboard to each place below. Every channel is approved by someone other than you before it goes live.
2 of 2 channels approved. Live since 11 Oct 2026.
Consume
Approved
Live
Consume · Shown in Signal Harbor
Set up by Mei Chen. Approved by Daniel Foo, 11 Oct 2026.
Preview
Email digest
Approved
Email digest · Every month at 07:00 · to family.council@evergreen-office.example.test
Set up by Mei Chen. Approved by Daniel Foo, 11 Oct 2026.
Email digest is not set up yet. Ask your administrator. Messages for it are kept in the outbox below.
Preview
What people see in Consume
Consume · Lumen, Signal Harbor · preview of the post
Portfolio return in the latest month, EG-LIQUID only: portfolio return 0.0417
Portfolio return in the latest month, EG-LIQUID only: portfolio return 0.0417
Liquid portfolio return against benchmark return, per month: month 2026-01-01T00:00:00, series Benchmark return, value -0.0070
Portfolio return by portfolio code in the latest month: portfolio code EG-B-OPS, portfolio return -0.4434
Portfolio return by benchmark code in the latest month: benchmark code EG-OFFICE-BLEND, portfolio return -0.8715
Portfolio return: month 2026-09-01, value -0.8715
Open the dashboard
Where is this from?
Email digest subject
Lumen's starter dashboard: portfolio return 0.0417
Preview. Names are masked when a message leaves Signal Harbor.
See a Slack thread in Consume
Nothing is sent when you press Go live. Names and figures leave only through the gateway, and only to places your administrator has set up.
See the project summary

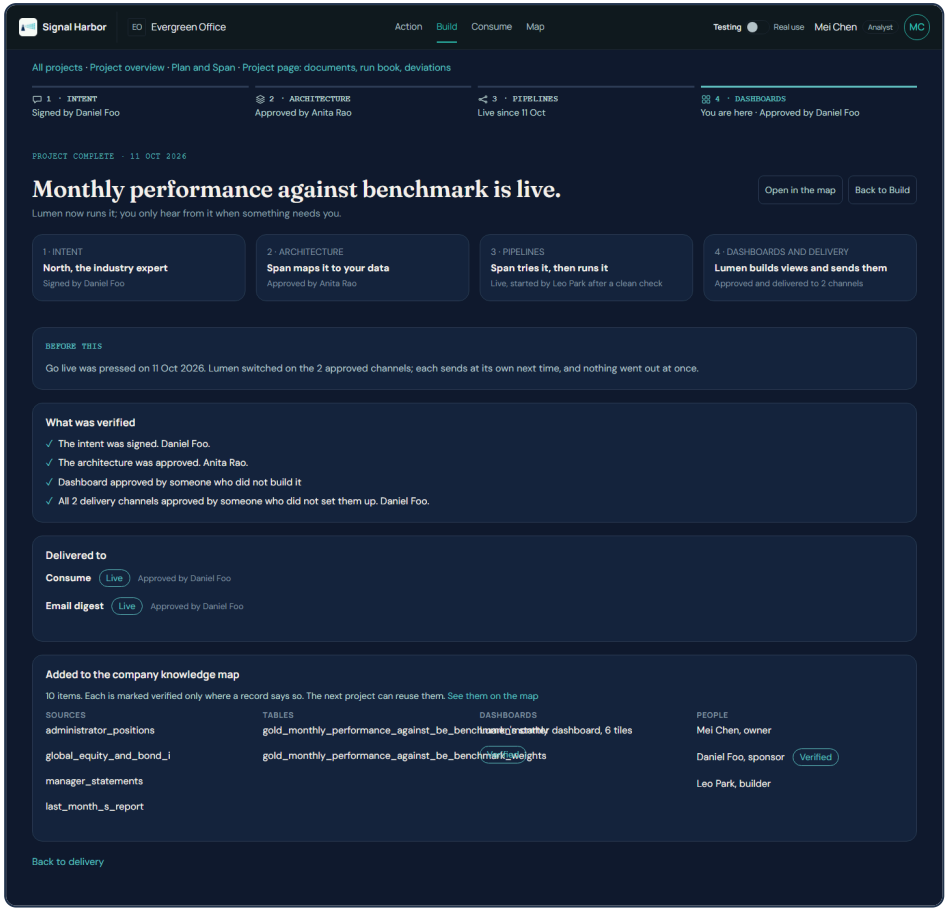

Signal Harbor
ED Evergreen Office
Action Build Consume Map
Testing Real use Mei Chen Analyst MC
All projects · Project overview · Plan and Span · Project page: documents, run book, deviations
1 · INTENT
Signed by Daniel Foo
2 · ARCHITECTURE
Approved by Anita Rao
3 · PIPELINES
Live since 11 Oct
4 · DASHBOARDS
You are here · Approved by Daniel Foo
Choose where it goes
Monthly performance against benchmark. Lumen sends the approved dashboard to each place below. Every channel is approved by someone other than you before it goes live.
Live. Nothing was sent just now; the first message goes at each channel's next time.
2 of 2 channels approved. Live since 11 Oct 2026.
Consume
Approved
Live
Consume · Shown in Signal Harbor
Set up by Mei Chen. Approved by Daniel Foo, 11 Oct 2026.
Preview
Stop
Email digest
Approved
Live
Email digest · Every month at 07:00 · to family.council@evergreen-office.example.test
Set up by Mei Chen. Approved by Daniel Foo, 11 Oct 2026.
Email digest is not set up yet. Ask your administrator. Messages for it are kept in the outbox below.
Preview
Stop
What people see in Email digest
Email digest · Lumen, Signal Harbor · preview of the post
Portfolio return in the latest month, EG-LIQUID only: portfolio return 0.0417
Portfolio return in the latest month, EG-LIQUID only: portfolio return 0.0417
Liquid portfolio return against benchmark return, per month: month 2026-01-01T00:00:00, series Benchmark return, value -0.0070
Portfolio return by portfolio code in the latest month: portfolio code EG-B-OPS, portfolio return -0.4434
Portfolio return by benchmark code in the latest month: benchmark code EG-OFFICE-BLEND, portfolio return -0.8715
Portfolio return: month 2026-09-01, value -0.8715
Open the dashboard
Where is this from?
Email digest subject
Lumen's starter dashboard: portfolio return 0.0417
Preview. Names are masked when a message leaves Signal Harbor.
See a Slack thread in Consume
Add a channel
Consume
Morning Note section
Slack channel
Microsoft Teams channel
Email digest
Threshold alert
Outgoing webhook
Dashboard
Lumen's starter dashboard
How often
Every day
At
07:00 am
Add as a draft
A draft is sent for approval to someone other than you.
Nothing is sent when you press Go live. Names and figures leave only through the gateway, and only to places your administrator has set up.
See the project summary

# 20 · Project complete

Mei Chen (business owner)

| | |
|---|---|
| Intended | The project's complete page records who signed, approved and delivered what, and adds the work to the company knowledge map. |
| What we did | Mei opened the project's complete page. |
| What happened | 'PROJECT COMPLETE - 11 OCT 2026. Monthly performance against benchmark is live. Lumen now runs it; you only hear from it when something needs you.' Intent signed by Daniel Foo; architecture approved by Anita Rao; pipelines 'Live, started by Leo Park after a clean check'; dashboards 'Approved and delivered to 2 channels'. Added to the company knowledge map: 10 items (4 sources, 2 gold tables, the dashboard, Mei as owner, Daniel as sponsor, Leo as builder), each verified only where a record says so. |
| Time taken | 20 s |
| If it goes wrong | Anita Rao (head of data) reopens a step; Daniel Foo stops delivery. |

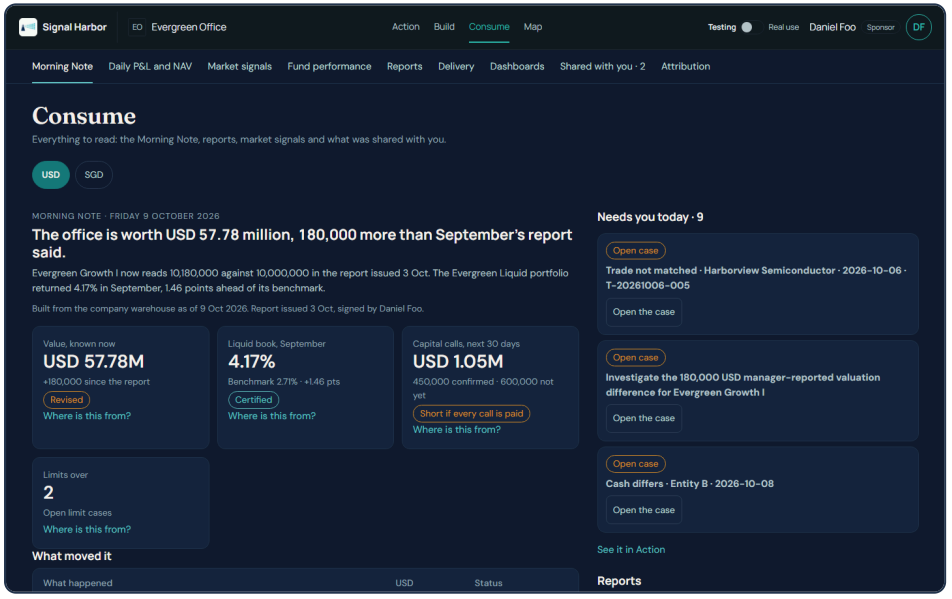

Signal Harbor
EO Evergreen Office
Action Build Consume Map
Testing Real use Mei Chen Analyst MC
All projects · Project overview · Plan and Span · Project page: documents, run book, deviations
1 · INTENT
Signed by Daniel Foo
2 · ARCHITECTURE
Approved by Anita Rao
3 · PIPELINES
Live since 11 Oct
4 · DASHBOARDS
You are here · Approved by Daniel Foo
PROJECT COMPLETE · 11 OCT 2026
Monthly performance against benchmark is live.
Open in the map
Back to Build
Lumen now runs it; you only hear from it when something needs you.
1 · INTENT
North, the industry expert
Signed by Daniel Foo
2 · ARCHITECTURE
Span maps it to your data
Approved by Anita Rao
3 · PIPELINES
Span tries it, then runs it
Live, started by Leo Park after a clean check
4 · DASHBOARDS AND DELIVERY
Lumen builds views and sends them
Approved and delivered to 2 channels
BEFORE THIS
Go live was pressed on 11 Oct 2026. Lumen switched on the 2 approved channels; each sends at its own next time, and nothing went out at once.
What was verified
✓ The intent was signed. Daniel Foo.
✓ The architecture was approved. Anita Rao.
✓ Dashboard approved by someone who did not build it
✓ All 2 delivery channels approved by someone who did not set them up. Daniel Foo.
Delivered to
Consume Live Approved by Daniel Foo

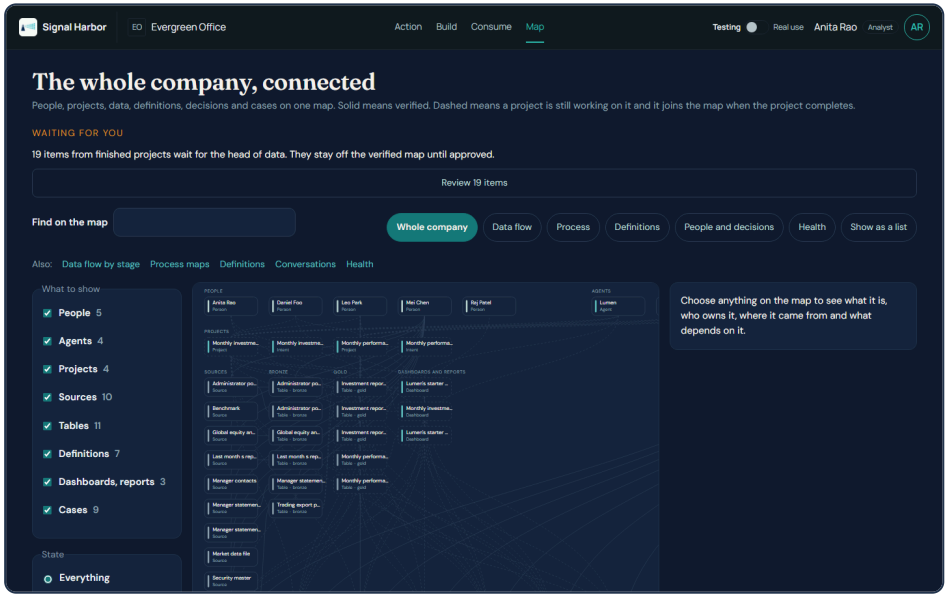

Signal Harbor
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Action Build Consume Map
Testing Real use Mei Chen Analyst MC
All projects · Project overview · Plan and Span · Project page: documents, run book, deviations
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3 · PIPELINES
Live since 11 Oct
4 · DASHBOARDS
You are here · Approved by Daniel Foo
PROJECT COMPLETE · 11 OCT 2026
Monthly performance against benchmark is live.
Open in the map
Back to Build
Lumen now runs it; you only hear from it when something needs you.
1 · INTENT
North, the industry expert
Signed by Daniel Foo
2 · ARCHITECTURE
Span maps it to your data
Approved by Anita Rao
3 · PIPELINES
Span tries it, then runs it
Live, started by Leo Park after a clean check
4 · DASHBOARDS AND DELIVERY
Lumen builds views and sends them
Approved and delivered to 2 channels
BEFORE THIS
Go live was pressed on 11 Oct 2026. Lumen switched on the 2 approved channels; each sends at its own next time, and nothing went out at once.
What was verified
✓ The intent was signed. Daniel Foo.
✓ The architecture was approved. Anita Rao.
✓ Dashboard approved by someone who did not build it
✓ All 2 delivery channels approved by someone who did not set them up. Daniel Foo.
Delivered to
Consume Live Approved by Daniel Foo
Email digest Live Approved by Daniel Foo
Added to the company knowledge map
10 items. Each is marked verified only where a record says so. The next project can reuse them. See them on the map
SOURCES
administrator_positions
global_equity_and_bond_i
manager_statements
last_month_s_report
TABLES
gold_monthly_performance_against_be_benchmark_weights
DASHBOARDS
Verified
PEOPLE
Mei Chen, owner
Daniel Foo, sponsor Verified
Leo Park, builder
Back to delivery

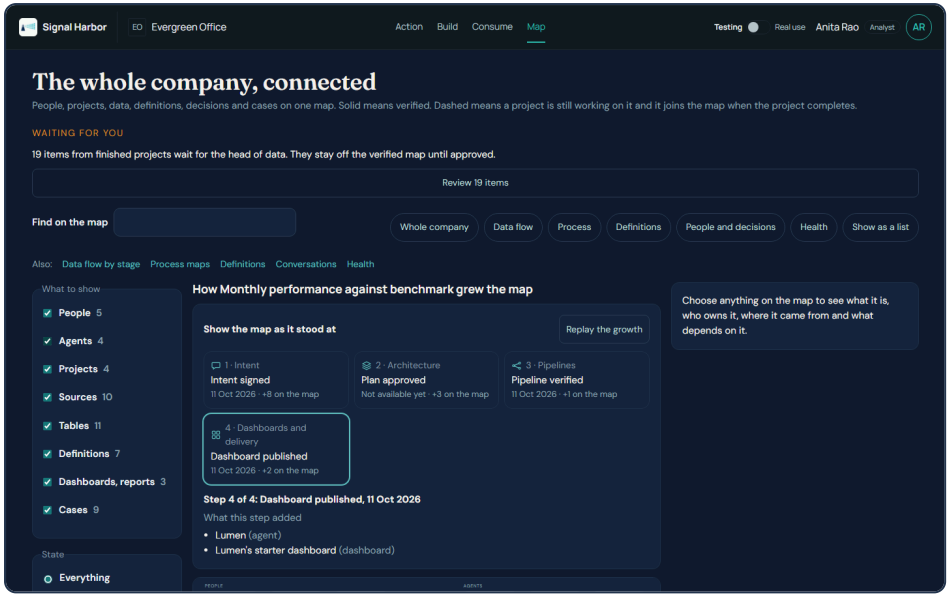
# 21 · Consume and the Map show the result

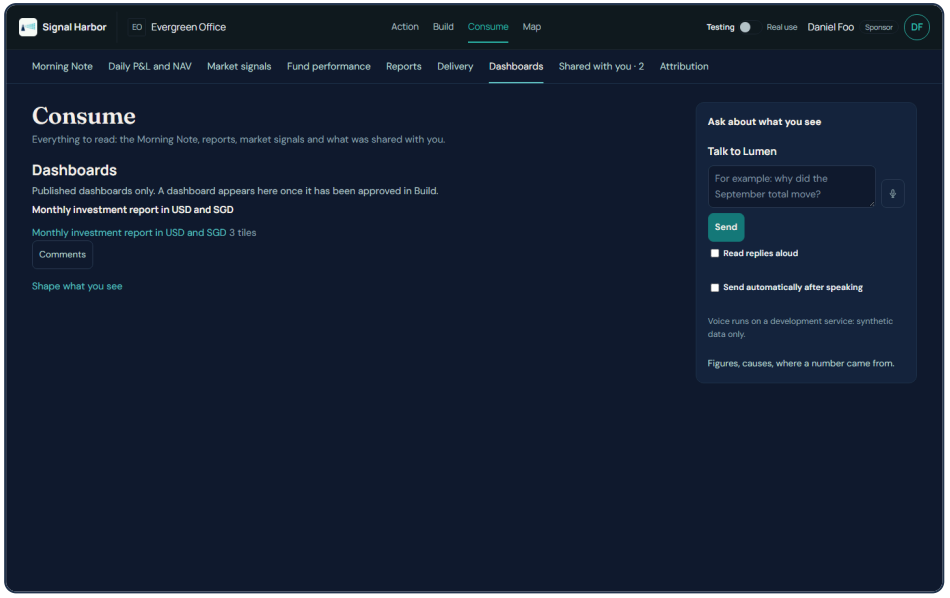
Daniel Foo (sponsor), Anita Rao (head of data)

| | |
|---|---|
| Intended | Daniel reads the result in Consume; Anita sees the project, its sources and tables on the company map. |
| What we did | Daniel opened Consume, its Dashboards list and Shared with you. Anita opened the Map and the project on it. |
| What happened | Consume's Morning Note reads 'The Evergreen Liquid portfolio returned 4.17% in September, 1.46 points ahead of its benchmark' with a Certified tile 'Liquid book, September 4.17%, Benchmark 2.71% +1.46 pts' (the same figures the pipeline matched). But the new dashboard is not listed under Consume > Dashboards or Shared with you, for Daniel or Mei, although its Consume channel is approved and live; only the earlier project's dashboard shows. The Map lists 'Monthly performance against benchmark - Live' beside the earlier project, with its sources (Administrator positions export, Global equity and bond index levels, Last month's report, Manager statements); 19 items from finished projects wait for the head of data before they join the verified map. |
| Time taken | 39 s |
| If it goes wrong | Mei checks the Consume channel on the delivery page; Anita reviews the 19 waiting map items. |




Signal Harbor
EO Evergreen Office
Action Build Consume Map
Testing Real use Anita Rao Analyst AR
The whole company, connected
People, projects, data, definitions, decisions and cases on one map. Solid means verified. Dashed means a project is still working on it and it joins the map when the project completes.
WAITING FOR YOU
19 items from finished projects wait for the head of data. They stay off the verified map until approved.
Review 19 items
Find on the map
Whole company
Data flow
Process
Definitions
People and decisions
Health
Show as a list
Also: Data flow by stage Process maps Definitions Conversations Health
What to show
People 5
Agents 4
Projects 4
Sources 10
Tables 11
Definitions 7
Dashboards, reports 3
Cases 9
State
Everything
How Monthly performance against benchmark grew the map
Show the map as it stood at
Replay the growth
1 · Intent
Intent signed
11 Oct 2026 · +8 on the map
2 · Architecture
Plan approved
Not available yet · +3 on the map
3 · Pipelines
Pipeline verified
11 Oct 2026 · +1 on the map
4 · Dashboards and delivery
Dashboard published
11 Oct 2026 · +2 on the map
Step 4 of 4: Dashboard published, 11 Oct 2026
What this step added
Lumen (agent)
Lumen's starter dashboard (dashboard)
Choose anything on the map to see what it is, who owns it, where it came from and what depends on it.


Signal Harbor
EO Evergreen Office
Action Build Consume Map
Testing Real use Daniel Foo Sponsor DF
Morning Note
Daily P&L and NAV
Market signals
Fund performance
Reports
Delivery
Dashboards
Shared with you · 2
Attribution
Consume
Everything to read: the Morning Note, reports, market signals and what was shared with you.
Dashboards
Published dashboards only. A dashboard appears here once it has been approved in Build.
Monthly investment report in USD and SGD
Monthly investment report in USD and SGD 3 tiles
Comments
Shape what you see
Ask about what you see
Talk to Lumen
For example: why did the September total move?
Send
Read replies aloud
Send automatically after speaking
Voice runs on a development service: synthetic data only.
Figures, causes, where a number came from.

# Summary

In the product, from Mei's first sentence to North (21:46 Singapore time) to delivery live (about 22:31): about 45 minutes, of which about 10 minutes were the conversation with North. The time recorded inside the 21 steps adds up to about 25 minutes; the rest was handing over between people. Each person acted only at their own points; the agents drafted, ran and checked in between. The whole walkthrough took about 63 minutes of wall-clock time: rebuilding the product screens, seeding the environment, making the sources and diagnosing the defects listed below.

## Touch points per person

| Person | Steps acted in |
|---|---|
| Leo Park | 11 |
| Mei Chen | 9 |
| Daniel Foo | 4 |
| Anita Rao | 3 |

## What the agents did, and what people did

| Agents | People |
|---|---|
| • North: 13 model calls; asked one question per turn, drafted the intent sections, the three figure definitions and the 5-step process map, wrote the closing summary<br>• Span the architect: read the signed intent (never asked again for sources, channels or definitions), asked 4 questions with defaults, profiled two samples, drafted the design pack (2 gold, 5 silver tables, 19 mapping rows, benchmark block), versioned the correction and showed the comparison<br>• Span the builder: connected 4 sources, ran on a copy by itself at approval (6 tables, 60 gold rows), read the uploaded report and compared 6 lines (all matched), reran pack version 2 by itself and rechecked | • Mei Chen: 10 messages to North; hand-over; asked for one tile; sent the dashboard for approval; set up two delivery channels and pressed Go live<br>• Daniel Foo: signed the intent; approved the dashboard and both channels<br>• Anita Rao: approved pack version 1, its plan, pack version 2 and its plan<br>• Leo Park: confirmed 4 sources, answered 4 questions, gave 4 samples and marked 6 fields, settled 2 mapping |

- Lumen: proposed a starter dashboard with 6 widgets from the signed intent; listed what it checked for the approver; wrote the delivery previews

rows, corrected 4 definitions, uploaded the benchmark report, pressed Go live

## Model use

Testing mode throughout (Real use never switched on). 26 model calls, about 198,000 tokens, 0.0106 USD in total from the product's cost log (North 13 calls, 0.0066 USD; the rest Lumen and summaries). Budget was under 1 USD.

# Honest findings

1. High **Mei's specific tile could not be added**. Asking Lumen for a tile while the starter is open puts the proposal on a new hidden personal 'My dashboard' (three were made), so Lumen says it waits for Accept but no Accept appears. The widget library's Figure takes the first column (portfolio return) with no column choice; selecting a tile to change it did not register; a later Lumen reply went off topic. The difference tile was not added.
2. High **The delivered dashboard is missing from Consume**. Approved and delivered to a live Consume channel, yet not listed under Consume > Dashboards or Shared with you for Daniel or Mei. Consume shows the same figures only through the seeded Morning Note.
3. High **North reversed a signed definition**. The 'Return difference' figure was drafted as 'difference in basis points (benchmark - portfolio)'; Mei said portfolio minus benchmark in percentage points. The sign page did not flag it and Daniel signed it. The pack computes portfolio minus benchmark, so the signed text and the build disagree (noted in pack version 2).
4. High **Lumen's starter tiles are partly wrong**. The same 4.17% tile twice; no tile for the benchmark return or the difference although the intent names both; by-portfolio and by-benchmark-code bars at about -50% and -90% (probably values added up across months or portfolios); trend axis in fractions next to tiles in percent; white number boxes on the dark theme. The email preview carries the same raw fractions (0.0417, -0.8715).
5. Medium **Approval panel says the numbers were not checked**. 'What Lumen checked' shows 'No reviewed run is on record' and 'No signed benchmark version is on record' after the builder's check matched the report and with EG-OFFICE-BLEND version 1 signed.
6. Medium **The gold tables do not read this month's files**. The pack's gold tables read the company model (positions, market observations, benchmark legs), not the uploaded Excel and CSV: both stayed 'Sample kept; map it before it feeds a table' and were settled in words. Bronze says 'no mapping is approved yet'; checks 0 of 1 after the run. The cycle proves the calculation on data already in the warehouse, not the ingestion of these files.
7. Medium **No place to land this month's further PDFs**. Pipelines has no upload; the Architecture sample takes one PDF per source; the intake page takes only Excel or CSV. The three other manager statements were not landed. The statement count reads '104 on file for the company', not this project.
8. Medium **PDF tables cannot be read**. The PDF reader puts each table cell on its own line, so a statement or report laid out as a two-column table cannot be marked. The walkthrough's first PDFs used tables and were re-issued with one 'label: value' line each.
9. Medium **North needed prompting to finish**. With every required section filled North did not mark the intent ready; it asked for the version 2 weights (not needed) and wrote the closing summary only when Mei asked. The summary listed each figure as 'no source yet', kept 'who may see this' as unclear after Mei named four people, and cut the goal mid-word ('so the council can de').

10. Low **Grain cannot be edited on screen**. The few questions offers only Accept for grain (Span proposed 'monthly by entity' for a per-portfolio figure); the change had to be entered outside the screen.
11. Low **Pack version 1 had copied definitions**. portfolio_code and benchmark_code carried a return KPI's definition text; the Definitions panel lists 2 of the 3 signed figures. Corrected in pack version 2.
12. Low **Two approvals for the head of data**. Anita approves the pack and then the plan drawn from it, where the agreed flow has one approval.
13. Low **Send-back loop not exercised**. The first run matched the report (4.1667 / 2.7072 / 1.4595 against 4.17 / 2.71 / 1.46), so 'Send back to the architect' was never offered. Pack version 2 came from Leo's own correction, and the builder reran by itself.
14. Low **Nothing sent at go-live**. Delivery's outbox stays empty after Go live (the first email waits for the schedule), so no sent message could be shown.
15. Low **Truncated names**. Gold tables are named 'gold_monthly_performance_against_be_...' and a source 'global_equity_and_bond_i' on Pipelines.